

Figure 13.5. Approximate solution of $\cos x = x$. The "spiral" approaches the point of intersection of $y = \cos x$ and $y = x$.

and we have established that the sequence is Cauchy. Since X is complete, there exists $x \in X$ such that $x_n \to x$. By Lemma 13.73 this point x is a fixed point of A. The uniqueness was given in Lemma 13.72. ■

Observe that the proof of Theorem 13.75 also provides a practical method for approximating the solution of an equation of the form $A(x) = x$. This method is often called the method of *successive approximations*.

We can choose x_0 to be *any* point in X. Then the sequence $\{A^n(x_0)\}$ converges to the unique solution of the equation $A(x) = x$. Indeed the inequality (10) shows that the convergence is as fast as some geometric progression $\{C\alpha^n\}$.

Example 13.76: Let us solve the equation

$$\cos x = x.$$

Ordinary calculus techniques should convince you that there is a unique solution, but what practical method would offer an approximate answer? No algebraic methods or trigonometric identities will lead to a solution. Instead let us interpret this as the requirement to find the fixed point of the function

$$f(x) = \cos x.$$

On the interval $[1/2, 1]$ this is evidently a contraction. Thus if we start with any value $x_0 \in [1/2, 1]$ and follow the sequence

$$x_n = \cos x_{n-1}$$

for $n = 1, 2, 3, \ldots$ the proof of Theorem 13.75 shows that this sequence must converge to that fixed point. Indeed this process can be tried on any calculator: Start by entering the number 1 (say) and then repeatedly press the cosine key. You will see the numbers approach 0.73908513 quite quickly. It is instructive to see what is happening graphically. (See Fig. 13.5.) ◀

Example 13.77: Let $X = C[0, 1/2]$ furnished, as usual, with the sup metric,

$$d(f_1, f_2) = \max_{x \in [0, \frac{1}{2}]} |f_1(x) - f_2(x)|.$$

Define a function A on X by

$$(A(f))(x) = g(x) = \int_0^x f(t)\,dt, \quad 0 \le x \le \frac{1}{2}.$$

Since g is continuous on $[0,1/2]$, $g \in X$ and $A : X \to X$.

For $f_1, f_2 \in X$, let $g_1 = A(f_1)$ and $g_2 = A(f_2)$. Then

$$\begin{aligned} d(g_1,g_2) &= \max_{x\in[0,\frac{1}{2}]} |g_1(x) - g_2(x)| \\ &= \max_{x\in[0,\frac{1}{2}]} \left| \int_0^x [f_1(t) - f_2(t)]\,dt \right| \\ &\le \max_{x\in[0,\frac{1}{2}]} \int_0^x |f_1(t) - f_2(t)|\,dt \\ &\le \max_{x\in[0,\frac{1}{2}]} x \max_{t\in[0,\frac{1}{2}]} |f_1(t) - f_2(t)| \\ &\le \frac{1}{2} d(f_1,f_2). \end{aligned}$$

Thus A is a contraction map.

From Theorem 13.75 we can conclude that there is a unique function $f \in \mathcal{C}[0,1/2]$ such that $A(f) = f$, that is, such that

$$f(x) = \int_0^x f(t)\,dt \text{ for all } x \in [0,1/2].$$

Since the zero function, $f(x) = 0$ for all $x \in [0,1/2]$, satisfies this and the solution is unique, we see that the zero function is the only solution. Had we not observed this, we could have tried the method of successive approximations.

Starting with $f(t) = 1$, for example, we obtain the sequence

$$t, \frac{t^2}{2}, \frac{t^3}{6}, \ldots, \frac{t^n}{n!}, \ldots$$

which converges in the space $\mathcal{C}[0,1/2]$ (that is, it converges uniformly) to the fixed point $f \equiv 0$. ◀

We shall see a variety of applications of Theorem 13.75 in the next section.

Exercises

13.9.1 Prove that if $f : \mathbb{R} \to \mathbb{R}$ is continuous and satisfies the Lipschitz condition

$$|f(x) - f(y)| \le \alpha|x - y| \text{ for all } x, y \in \mathbb{R} \text{ with } 0 < \alpha < 1,$$

then f has a unique fixed point.

13.9.2 Let $f(x) = x + 1/x$. Choose a point $x_0 \in [2,\infty)$ and observe what happens to the sequence $\{x_k\} = \{f^k(x_0)\}$. Where would the proof of Theorem 13.75 break down if we try to apply it to this mapping, f?

13.9.3 Show that the function $f(x) = \cos x$ is a contraction mapping on $[1/2,1]$ but is not a contraction map on $[0,\pi]$.

13.9.4 Show that the function $f(x) = \cos x$ is not a contraction mapping on $\mathbb{R}$ but that some iterate of f is a contraction map on $\mathbb{R}$. Can Theorem 13.75 be applied in this case?

13.9.5 Prove that a contraction map on a metric space is continuous.

13.9.6 Show that we cannot drop the requirement that X is complete in Theorem 13.75 without rendering the conclusion false.

13.9.7 A careless student claims the function e^x is a fixed point of the map in Example 13.77. Take this function as the initial approximation for a sequence of successive approximations. Obtain also the sequence of successive approximations when the initial function f_0 is $\sin x$.

13.9.8 Define a function $A : \mathcal{C}[0,1] \to \mathcal{C}[0,1]$ by

$$(A(f))(x) = g(x) = \int_0^x f(t)\,dt, \quad 0 \le x \le 1.$$

(a) Is A a contraction?

(b) Is A^2 a contraction?

(c) Does A have a unique fixed point?

SEE NOTE 348

13.9.9 Let A and B be contraction maps on a complete metric space (X,d) and suppose that

$$d(A(x),B(x)) < \varepsilon$$

for all $x \in X$. If a is the fixed point of A and b is the fixed point of B find an estimate for $d(a,b)$.

SEE NOTE 349

13.9.10 Let A be a contraction map on a complete metric space (X,d) such that $d(A(x),A(y)) \le \alpha d(x,y)$ for all $x, y \in X$. Show that the "rate of convergence" of the sequence of iterates $\{A^n(x_0)\}$ in Theorem 13.75 to its limit x can be estimated by

$$d(A^n(x_0),x) \le \frac{\alpha^n d(x_0,A(x_0))}{1-\alpha}.$$

13.9.11 Let $\{A_n\}$ be a sequence of contraction maps on a complete metric space (X,d) such that

$$d(A_n(x),A_n(y)) \le \alpha d(x,y) \tag{11}$$

for all $x, y \in X$ and $n \in \mathbb{N}$ and some $\alpha < 1$. Suppose that $\{A_n(x)\}$ converges to $A(x)$ for each $x \in X$. Show that A is a contraction mapping on X and that its fixed point can be computed as

$$a = \lim_{n\to\infty} a_n,$$

where a_n is the fixed point of the contraction map A_n.

13.9.12 In Hilbert space ℓ_2 consider the mappings $A_n : \ell_2 \to \ell_2$ defined so that if $(x_1,x_2,x_3,\dots) \in \ell_2$, then $A_n((x_1,x_2,x_3,\dots))$ is that element of ℓ_2 with all zero entries except for the number

$$\frac{1}{n} + \frac{n-1}{n} x_n$$

in the nth position. Show that A_n is a contraction mapping with a unique fixed point $a_n \in \ell_2$ but that the sequence $\{a_n\}$ does not converge in ℓ_2. Conclude from this example that the condition (11) in Exercise 13.9.11 cannot be dropped.

13.9.13 Let $\{A_n\}$ be a sequence of mappings of a complete metric space (X,d) into itself such that $\{A_n(x)\}$ converges uniformly to $A(x)$. Suppose further that A is a contraction mapping and that for each n there is at least one fixed point a_n of the mapping A_n. Show that the fixed point of A can be computed as

$$a = \lim_{n\to\infty} a_n.$$

SEE NOTE 350

13.10 Applications of Contraction Maps (I)

✂ *Enrichment section. May be omitted.*

A variety of problems in analysis involve the solving of some sort of equations. The equations could be ones involving numbers, or n-tuples of numbers, or sequences or functions or various other mathematical objects. Often such a problem can be cast in the following form: We observe that a certain operator arises naturally in connection with the problem, and that a solution to the problem is representable as a fixed point of the operator.

Let's look at a few examples. We will not consider the solutions until the next section. For now we only reinterpret the problem as a fixed point problem. We begin with a trivial example just to illustrate the approach.

Example 13.78: (A Simple Linear Equation) Consider the simple linear equation

$$ax = b, \quad (a \neq 0). \tag{12}$$

We can rewrite the equation as $x = (1-a)x + b$. Now consider the operator $A(x) = (1-a)x + b$. A solution to (12) is just a fixed point of the operator A. ◀

Example 13.79: (Systems of Linear Equations) Consider a system of linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned} \tag{13}$$

We can rewrite this system in the form

$$\begin{aligned} x_1 &= c_{11}x_1 + c_{12}x_2 + \cdots + c_{1n}x_n + b_1 \\ x_2 &= c_{21}x_1 + c_{22}x_2 + \cdots + c_{2n}x_n + b_2 \\ &\vdots \\ x_n &= c_{n1}x_1 + c_{n2}x_2 + \cdots + c_{nn}x_n + b_n, \end{aligned} \tag{14}$$

where $c_{ij} = -a_{ij}$ if $j \neq i$, and $c_{ii} = 1 - a_{ii}$. We can then find solutions to the system (13) by solving the equivalent system (14).

It is now easy to view this problem in terms of a fixed point of an operator: For $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, let $y = A(x)$, where $y = (y_1, \ldots, y_n)$ with

$$y_i = \sum_{j=1}^{n} c_{ij} x_j + b_i.$$

Thus $A : \mathbb{R}^n \to \mathbb{R}^n$. A solution of (14) is just a fixed point of the operator A. ◀

Example 13.80: (Infinite Systems of Linear Equations) The preceding ideas can be applied to infinite systems of linear equations. In the late nineteenth century, a number of authors considered such systems arising, for example, in studies of algebraic equations and celestial mechanics. Curiously, the first person to encounter an infinite system of linear equations was Joseph Fourier (1768–1830) in his classic 1822 study. His methods were simple, but unjustified. After that, the subject received no further attention for another half century.

Suppose we have an infinite system of equations of the form

$$x_i = \sum_{j=1}^{\infty} c_{ij} x_j + b_i \qquad (i = 1, 2, 3, \ldots). \tag{15}$$

We seek a sequence $x = \{x_i\}$ that satisfies (15). As in Example 13.79, we can consider the operator A defined by $y = A(x)$, where $x = \{x_i\}$, $y = \{y_i\}$ and $y_i = \sum_{j=1}^{\infty} c_{ij} x_j + b_i$. As before, a solution to (15) could be viewed as a fixed point of the operator A, but here we must be a bit more careful. In Example 13.79, it was clear that the domain and range of the operator was $\mathbb{R}^n$. Here we have a number of sequence spaces that we might consider (e.g. ℓ_1, ℓ_2, ℓ_∞). (These spaces were described in Examples 13.7–13.9.) We continue our discussion of this example in the next section. ◀

Example 13.81: (Fredholm's Integral Equation) You may be familiar with certain "boundary value problems" from mathematical physics, such as the Dirichlet problem and the Neumann problem. Our next example involves an integral equation that together with its relatives plays a role in dealing with such problems.

Consider the equation

$$f(x) = \lambda \int_a^b K(x,y) f(y)\, dy + \phi(x), \tag{16}$$

where $\lambda \in \mathbb{R}$, ϕ is continuous on $[a,b]$, and K is continuous on $[a,b] \times [a,b]$. We seek a function $f \in \mathcal{C}[a,b]$ that satisfies (16). It is natural to consider the operator A defined on $\mathcal{C}[a,b]$ by $A(f) = g$, where

$$(A(f))(x) = g(x) = \lambda \int_a^b K(x,y) f(y)\, dy + \phi(x).$$

A fixed point of this operator provides a solution to (16). As in Example 13.80, we must be precise about the range of this operator. We continue our discussion in the next section. ◀

Example 13.82: (Differential Equations) Let D be an open set in $\mathbb{R}^2$ and let $f : D \to \mathbb{R}$. We wish to find a local solution to the differential equation

$$\frac{dy}{dx} = f(x,y), \qquad y(x_0) = y_0.$$

For a specific and familiar type of example we could ask for the solution of

$$\frac{dy}{dx} = x^2 \sin xy + ye^{xy}$$

that "passes through" the point $(0,0)$.

Can we be sure such a solution exists? Naturally, some conditions on f must be imposed. For the moment, let's just see how to cast the problem in terms of operators.

We begin by reformulating the problem in terms of an integral equation. We seek a function ϕ such that

$$\phi'(x) = f(x,\phi(x)) \text{ and } \phi(x_0) = y_0$$

on some open interval containing x_0. We can recast the problem as seeking ϕ such that

$$\phi(x) = y_0 + \int_{x_0}^{x} f(t,\phi(t))\,dt.$$

An appropriate operator A of the form

$$(A(\phi))(x) = y_0 + \int_{x_0}^{x} f(t,\phi(t))\,dt$$

would be natural for this problem. We must be careful to impose conditions on f that allow us to define A on an appropriate function space. We shall do this in the next section. ◀

We have presented several historically important examples of problems whose solutions involve fixed points of operators. In each case, appropriate restrictions on the class of sequences or functions to be considered allows us to use the Banach fixed point theorem (Theorem 13.75) to show that there is a fixed point—in fact a unique one—in the space under consideration. Each of the theorems we obtain in our next section addresses one of the examples mentioned in this section.

Exercises

13.10.1 What conditions on the values of a and b in Example 13.78 will force the function $A(x) = (1-a)x + b$ to be a contraction on $\mathbb{R}$? Examine the sequence of iterates in this case.

SEE NOTE 351

13.10.2 Apply the method of Example 13.79 to solve the system of two linear equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 &= b_1 \\ a_{21}x_1 + a_{22}x_2 &= b_2. \end{aligned}$$

What sufficient conditions on the numbers a_{11}, a_{12}, a_{21}, a_{22}, b_1, and b_2 will force the function chosen by this method to be a contraction on $\mathbb{R}^2$?

SEE NOTE 352

13.10.3 Prove this special version of the intermediate value theorem: If f is a differentiable function on the interval $[a,b]$ with $f(a) < 0 < f(b)$ and $0 < c \le f'(x) \le C$, then there is a unique solution of the equation $f(x) = 0$ that can be obtained by iterating the function

$$F(x) = x - \frac{f(x)}{C}.$$

SEE NOTE 353

13.11 Applications of Contraction Maps (II)

✂ *Enrichment section. May be omitted.*

In Section 13.10 we saw that solutions to various equations or systems of equations correspond to fixed points of operators associated with the equations. The fixed point theorem of Section 13.9 can sometimes be used to guarantee that there is a solution, in fact a unique one.

In this section we revisit each of the examples from Section 13.10. We obtain conditions under which the relevant operators are contraction maps. When they are, there will be unique solutions. And we can obtain the solutions by the method of successive approximations. Observe that in each case we must obtain a complete metric space on which the operator is a contraction. The conclusion of Theorem 13.75 will then apply in that space.

Example 13.83: (**Example 13.78 Revisited.**) A solution to the equation $ax = b$ is just a fixed point of the operator

$$A(x) = (1-a)x + b.$$

We are considering here the metric space $\mathbb{R}$ (with the usual metric). We find $A : \mathbb{R} \to \mathbb{R}$. In order to determine whether A is a contraction, we calculate for $x, y \in \mathbb{R}$,

$$\begin{aligned} d(A(x),A(y)) &= |A(x) - A(y)| = |(1-a)(x-y)| \\ &= |1-a||x-y| = |1-a|d(x,y). \end{aligned}$$

Thus A is a contraction if and only if $|1-a| < 1$. According to Theorem 13.75, the equation $ax = b$ will have a unique solution in $\mathbb{R}$ provided that $|1-a| < 1$, that is provided that $0 < a < 2$.

Students of elementary algebra know that the equation $ax = b$ has the unique solution $x = b/a$ provided that $a \neq 0$, no solution if $a = 0$ and $b \neq 0$, and every real number x as a solution if $a = b = 0$. The point here is that the application of the contraction mapping principle is available only under the restricted condition that $|1-a| < 1$. ◀

We presented this example partially as a warm-up and partially to emphasize that Theorem 13.75 provides a *sufficient* condition for a unique fixed point, not a *necessary* condition.

We continue with several examples of historically important problems to which the Banach fixed-point theorem can be applied. There are many other interesting applications. You can find one involving the space $\mathcal{K}$ of compact subsets of the plane furnished with the Hausdorff metric.[3] It pertains to the technique of "fractal image compression" that is useful for encoding and storing graphic images in computers. Another application, given as Exercise 13.11.3, provides a proof of an implicit function theorem.

13.11.1 Systems of Equations (Example 13.79 Revisited)

✂ *Enrichment section. May be omitted.*

[3]There is material on the "collage theorem" in Bruckner, Bruckner, and Thomson, op. cit., p. 432.

We saw in Example 13.79 that the system of equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= b_n \end{aligned} \tag{17}$$

can be solved by finding a fixed point of the operator $A : \mathbb{R}^n \to \mathbb{R}^n$ defined by $y = A(x)$, if

$$y_i = \sum_{j=1}^{n} c_{ij}x_j + b_i \quad \text{with} \quad x = (x_1, \ldots, x_n),\ y = (y_1, \ldots, y_n).$$

In order for us to apply Theorem 13.75 we must specify the metric on $\mathbb{R}^n$—whether A is a contraction will depend on our choice of that metric. Let us choose the metric

$$d_\infty(x, y) = \max_{1 \le i \le n} |x_i - y_i|.$$

(See Exercise 13.2.6.) In that case, let $x, x^* \in \mathbb{R}^n$ with $y = A(x)$, $y^* = A(x^*)$ and compute

$$\begin{aligned} d(A(x), A(x^*)) &= d(y, y^*) = \max_i |y_i - y_i^*| \\ &= \max_i \left| \sum_j c_{ij}(x_j - x_j^*) \right| \\ &\le \max_i \sum_j |c_{ij}||x_j - x_j^*| \\ &\le (\max_i \sum_j |c_{ij}|)(\max_j |x_j - x_j^*|) \\ &\le \max_i \sum_j |c_{ij}| d(x, x^*). \end{aligned}$$

Thus A will be a contraction map if there is a number α with

$$\sum_j |c_{ij}| \le \alpha < 1 \text{ for all } i = 1, \ldots, n.$$

Observe that Theorem 13.75 guarantees a unique solution in the metric space $(\mathbb{R}^n, d_\infty)$ whenever there exists α such that

$$\sum_j |c_{ij}| \le \alpha < 1 \text{ for all } i = 1, \ldots, n.$$

Exercise 13.11.1 provides a different condition involving column sums in place of row sums.

13.11.2 Infinite Systems (Example 13.80 revisited)

✂ *Enrichment section. May be omitted.*

Here we are considering the infinite system of equations

$$x_i = \sum_{j=1}^{\infty} c_{ij}x_j + b_i \qquad (i = 1, 2, 3, \ldots). \tag{18}$$

This system leads to the operator $y = A(x)$, where

$$x = \{x_i\}, \;\; y = \{y_i\} \;\text{ and }\; y_i = \sum_{j=1}^{\infty} c_{ij}x_j + b_i. \tag{19}$$

If we wish to find conditions under which A is a contraction map, we must, as before, indicate which metric space we have under consideration. And here this decision is critical. Suppose, for example, we choose the space of bounded sequences known as m or as ℓ_∞ with the metric

$$d_\infty(x,y) = \sup_i(|x_i - y_i|).$$

(See Example 13.9.)

Since we wish A to map ℓ_∞ into itself, we impose the requirement that $\{b_i\} \in \ell_\infty$; that is, there exists $B < \infty$ such that

$$|b_i| < B \;\text{ for all }\; i \in \mathbb{N}. \tag{20}$$

Our work with the previous example suggests the limitation

$$\sum_{j=1}^{\infty} |c_{ij}| \leq \alpha < 1 \;\text{ for all }\; i \in \mathbb{N}. \tag{21}$$

Suppose then that the system (18) satisfies conditions (20) and (21) and that A is defined by (19). We wish to show that A is a contraction map on ℓ_∞.

We first verify that A maps ℓ_∞ into ℓ_∞. For $x = \{x_1, x_2, \dots\}$, an element of the space ℓ_∞, write $\|x\|_\infty = \sup_j |x_j|$. From (18), (20), and (21) we find that

$$|y_i| \leq \sum_{j=1}^{\infty} |c_{ij}|\|x\|_\infty + |b_i| \leq \alpha\|x\|_\infty + B. \tag{22}$$

Since (22) is valid for all $i \in \mathbb{N}$, we see that

$$\|A(x)\|_\infty = \|y\|_\infty = \sup_i |y_i| \leq \alpha \sup_j |x_j| + B,$$

so $A(x) \in \ell_\infty$. Thus A maps ℓ_∞ into ℓ_∞.

We next show that A is a contraction map. Let $x, x^* \in \ell_\infty$, $y = A(x)$ and $y^* = A(x^*)$. Then

$$y_i^* - y_i = \sum_{j=1}^{\infty} c_{ij}(x_j^* - x_j).$$

Using (21), we conclude that $|y_i^* - y_i| \leq \alpha\|x^* - x\|_\infty$, so

$$\|A(x^*) - A(x)\|_\infty = \sup_i |y_i^* - y_i| \leq \alpha\|x^* - x\|_\infty.$$

But this means that

$$d(A(x^*), A(x)) \leq \alpha d_\infty(x^*, x),$$

and we see that A is a contraction map.

We summarize this discussion as a theorem.

Theorem 13.84: *If the system of equations*

$$x_i = \sum_{j=1}^{\infty} c_{ij}x_j + b_i \;,\; i = 1, 2, 3, \dots$$

satisfies the two conditions

1. *There exists $B < \infty$ such that $|b_i| < B$ for $i = 1,2,\ldots$, and*
2. $\sum_{j=1}^{\infty} |c_{ij}| \leq \alpha < 1$ *for* $i = 1,2,\ldots$,

then this system has a unique solution in ℓ_∞.

Note. The conclusion of Theorem 13.84 guarantees a unique solution to the system of equations *in the space* ℓ_∞. Consider, for example, the system

$$x_1 = \frac{1}{2}x_2,\ x_2 = \frac{1}{2}x_3,\ x_3 = \frac{1}{2}x_4, \ldots\ .$$

For any $c \in \mathbb{R}$ the sequence $\{c, 2c, 4c, \ldots\}$ is a solution to this system. This does not contradict Theorem 13.84, however, since such a sequence is in ℓ_∞ if and only if $c = 0$.

13.11.3 Integral Equations (Example 13.81 revisited)

✂ *Enrichment section. May be omitted.*

Here we seek solutions to the integral equation

$$f(x) = \lambda \int_a^b K(x,y) f(y)\, dy + \phi(x) \tag{23}$$

The corresponding operator A is defined on $\mathcal{C}[a,b]$ by $A(f) = g$, where

$$(A(f))(x) = g(x) = \lambda \int_a^b K(x,y) f(y)\, dy + \phi(x).$$

We leave as Exercise 13.11.2 the fact that $A : \mathcal{C}[a,b] \to \mathcal{C}[a,b]$, that is, for each $f \in \mathcal{C}[a,b]$, $A(f) = g$ is also continuous on $[a,b]$.

We wish to find conditions under which A is a contraction map. Let $f_1, f_2 \in \mathcal{C}[a,b]$, let $g_1 = A(f_1)$, $g_2 = A(f_2)$, and let

$$M = \max\{|K(x,y)| : a \leq x \leq b, a \leq y \leq b\}.$$

(See Exercise 13.6.18.) Then

$$\begin{aligned} d(g_1,g_2) &= \max_{a \leq x \leq b} |g_1(x) - g_2(x)| \\ &\leq |\lambda| M \max_{a \leq x \leq b} |f_1(x) - f_2(x)| (b-a) \\ &\leq |\lambda| M (b-a) d(f_1,f_2). \end{aligned}$$

It follows that A is a contraction map provided that

$$|\lambda| < \frac{1}{M(b-a)}.$$

Thus, the method of successive approximations can be used to obtain the unique solution to (23) provided that $|\lambda|$ is not too big.

13.11.4 Picard's Theorem (Example 13.82 revisited)

✂ *Enrichment section. May be omitted.*

Our aim is to prove the classical theorem of Picard, which is an application of contraction mappings to a problem in differential equations. We first need a definition. You will recall what was meant by a Lipschitz condition for a function $f : \mathbb{R} \to \mathbb{R}$. We now extend this meaning to allow for real-valued functions of two variables.

Definition 13.85: Let D be an open set in $\mathbb{R}^2$ and let $f : D \to \mathbb{R}$. We say that f satisfies a *Lipschitz condition in y on D*, with *Lipschitz constant M* if

$$|f(x,y_2) - f(x,y_1)| \leq M|y_2 - y_1| \tag{24}$$

whenever (x,y_1) and (x,y_2) are in D.

Theorem 13.86 (Picard) *Let f be a continuous function on D and satisfy a Lipschitz condition in y on D with Lipschitz constant M, and let $(x_0,y_0) \in D$. Then there exists $\delta > 0$ such that the differential equation*

$$\frac{dy}{dx} = f(x,y) \tag{25}$$

has a unique solution $y = \phi(x)$ in the interval $[x_0 - \delta, x_0 + \delta]$ for which $\phi(x_0) = y_0$.

Proof. As we indicated in Example 13.82, we formulate our problem in terms of the integral equation

$$\phi(x) = y_0 + \int_{x_0}^{x} f(t,\phi(t))\,dt, \tag{26}$$

which is to be valid on the interval $[x_0 - \delta, x_0 + \delta]$. Since D is open, there exists a closed sphere S centered at (x_0,y_0) and contained entirely inside the open set D. Since f is continuous on D, we may let K be the maximum of $|f(x,y)|$ for points (x,y) in the sphere S.

Now we use the Lipschitz constant M [defined as in the inequality (24)]. Choose $\delta > 0$ such that $\delta < 1/M$ and so that every point (x,y) with

$$|x - x_0| \leq \delta \text{ and } |y - y_0| \leq K\delta$$

belongs to the sphere S. We then arrive at a rectangle

$$N = [x_0 - \delta, x_0 + \delta] \times [y_0 - K\delta, y_0 + K\delta]$$

that lies entirely inside the open set D (Fig. 13.6).

Consider now the map A defined so that

$$(A(\phi))(x) = \psi(x) = y_0 + \int_{x_0}^{x} f(t,\phi(t))\,dt$$

for $x_0 - \delta \leq x \leq x_0 + \delta$. Note that this may not defined as a map on the space $\mathcal{C}[x_0 - \delta, x_0 + \delta]$ since the values of the function $\phi(t)$ might not allow us to conclude that $(t,\phi(t))$ is in D so that $f(t,\phi(t))$ is defined. Thus we pass to a subspace.

Let $\mathcal{C}_1$ consist of those members of $\mathcal{C}[x_0 - \delta, x_0 + \delta]$ that satisfy $\phi(x_0) = y_0$ and for which

$$|\phi(x) - y_0| \leq K\delta \quad \text{for all } x \in [x_0 - \delta, x_0 + \delta].$$

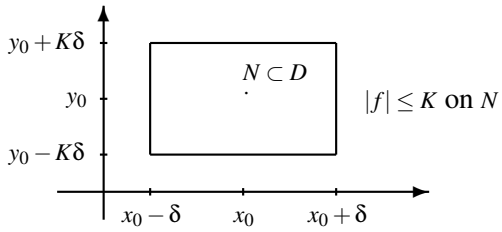


Figure 13.6. Choice of δ in the proof of Picard's theorem.

Then $\mathcal{C}_1$ is a closed subspace of the space $\mathcal{C}[x_0-\delta, x_0+\delta]$ and is therefore complete by Theorem 13.64. We show that the operator A defined previously maps $\mathcal{C}_1$ into itself. Let $x \in [x_0-\delta, x_0+\delta]$ and suppose $\phi \in \mathcal{C}_1$. Then

$$\begin{aligned} |\psi(x) - y_0| &= \left| \int_{x_0}^{x} f(t, \phi(t))\, dt \right| \le \int_{x_0}^{x} |f(t, \phi(t))|\, dt \\ &\le K|x - x_0| \le K\delta, \end{aligned}$$

so $\psi = A(\phi) \in \mathcal{C}_1$, and $A : \mathcal{C}_1 \to \mathcal{C}_1$.

We show that A is a contraction map on $\mathcal{C}_1$. To verify the contraction condition, let $\phi_1, \phi_2 \in \mathcal{C}_1$, let $\psi_1 = A(\phi_1)$ and let $\psi_2 = A(\phi_2)$. Then for all $x \in [x_0-\delta, x_0+\delta]$,

$$\begin{aligned} |\psi_1(x) - \psi_2(x)| &\le \int_{x_0}^{x} |f(t, \phi_1(t)) - f(t, \phi_2(t))|\, dt \\ &\le M\delta \max_x |\phi_1(x) - \phi_2(x)|. \end{aligned} \tag{27}$$

The last inequality is a consequence of the Lipschitz condition on f and the inequality $|x - x_0| \le \delta$. Now (27) is valid for all x in the interval $[x_0-\delta, x_0+\delta]$, so

$$d(\psi_1, \psi_2) \le M\delta d(\phi_1, \phi_2).$$

Since $M\delta < 1$, A is a contraction map, so the equation $\phi = A(\phi)$ has a unique solution in $\mathcal{C}_1$. In other words, the equation (26) and hence the equivalent equation (25) have unique local solutions. ■

Exercises

13.11.1 Show that the operator A of Example 13.79 revisited is a contraction on $(\mathbb{R}^n, d_1)$ if there exists α such that

$$\sum_{i=1}^{n} |c_{ij}| \le \alpha < 1 \text{ for all } j = 1, \ldots, n.$$

SEE NOTE 354

13.11.2 Show that the operator A of Example 13.81 revisited maps $\mathcal{C}[a,b]$ into itself.

13.11.3 Use Theorem 13.75 to prove the following form of the implicit function theorem. For other versions of such theorems and different proofs, see Section 12.6.

Theorem *Let $D = [a,b] \times \mathbb{R}$, and let $F : D \to \mathbb{R}$. Suppose that F is continuous on D and $\partial F/\partial y$ exists on D. If there exist positive real numbers α and β such that*

$$\alpha \leq \frac{\partial F}{\partial y} \leq \beta$$

on D, then there exists a unique function $f \in \mathcal{C}[a,b]$ such that

$$F(x, f(x)) = 0 \text{ for all } x \in [a,b].$$

That is, the equation $F(x,y) = 0$ can be solved uniquely for y as a continuous function of x on $[a,b]$.

SEE NOTE 355

13.12 Compactness

One of the goals of an abstract subject, such as the study of metric spaces, is to lift ideas and methods from our earlier studies into this abstract setting. We have already seen how the Cauchy criterion has entered the general subject of metric spaces. On the real line the Cauchy criterion is a necessary and sufficient condition for a sequence to converge. In our new theory we have defined a space to be *complete* precisely when it has this same property. Thus we can lift our methods from the real line into metric space theory. We verify that some space that we are studying is complete and then we have available to us the theory of Cauchy sequences.

In an similar way we wish now to lift our compactness arguments from Section 4.5 into metric space theory. You will recall that the Bolzano-Weierstrass and Heine-Borel theorems offered a powerful tool in the study of properties of continuous functions. On the real line these tools could be used in any closed and bounded set. They are not, however, available in every metric space since closed and bounded sets do not necessarily have these properties. Instead we shall define a set to be compact *precisely when the Bolzano-Weierstrass and Heine-Borel theorems are available.*

In Section 13.12.1 we take the Bolzano-Weierstrass property as our definition of compactness and derive some consequences. In Section 13.12.2 we show that some of our theorems from the elementary analysis of real functions can be extended to continuous functions defined on compact sets in a metric space. In Section 13.12.3 we show that the Heine-Borel property and the Bolzano-Weierstrass property are equivalent in any metric space so that either could have been taken as a definition of compactness. If you are intending to go on to even more abstract levels of generality, we should warn that there, in the subject of topology, these two concepts are not equivalent and that compactness is usually defined using the Heine-Borel property only.

13.12.1 The Bolzano-Weierstrass Property

Every bounded sequence of real numbers has a convergent subsequence. This is false in a general metric space.

Example 13.87: Let X be an infinite set furnished with the discrete metric. Let $\{x_n\}$ be any sequence of distinct elements of X. Then $\{x_n\}$ is bounded; indeed the diameter

of the whole space X is only 1 so every subset of X is bounded. But $\{x_n\}$ can have no convergent subsequence. ◀

Thus a special property of closed and bounded subsets of $\mathbb{R}$, namely the Bolzano-Weierstrass property, is false. This property asserts that if $K \subset \mathbb{R}$ is closed and bounded, then every sequence of points in K has a subsequence converging to a point in K. We have called closed and bounded subsets of $\mathbb{R}$ *compact*, but it was mainly this property of such sets that we needed. The natural thing to do in a general metric space is to recognize that "closed and bounded" no longer plays an important role but to turn the Bolzano-Weierstrass property itself into a definition of what it means to be compact.

Definition 13.88: A set K in a metric space is said to be *compact* provided that set has the Bolzano-Weierstrass property, namely that every sequence of points in K has a subsequence converging to a point in K.

If (X,d) is a metric space and the set X itself is compact (i.e., X has the Bolzano-Weierstrass property), then we say that the metric space is compact.

Let us begin by noting some properties that every compact set must have. The first has been left as Exercise 13.12.2.

Theorem 13.89: *Compactness is a topological property.*

Theorem 13.90: *Let K be a compact set in a metric space (X,d). Then*

1. *K is closed.*
2. *K is bounded.*
3. *K is complete.*
4. *K is separable.*

Proof. If K has the Bolzano-Weierstrass property, then every sequence of points in K that converges to a point z must have that point $z \in K$. Consequently, K must be closed.

Further, K must be bounded. If not, then for every integer N and any fixed point x_0 there must be points $x_N \in K$ with $d(x_0,x_N) > N$. This constructs a sequence that has every subsequence unbounded and, hence, could have no convergent subsequence. Thus if K has the Bolzano-Weierstrass property, K cannot be unbounded.

Further, K must be complete. If $\{x_n\}$ is a Cauchy sequence in K, then, by the Bolzano-Weierstrass property, there is a subsequence $\{x_{n_k}\}$ convergent to a point $z \in K$. But any Cauchy sequence with a convergent subsequence is itself convergent (Exercise 13.8.6).

Finally, let us show that K is separable. We can work just in the metric space (K,d). Suppose that K is not separable. Then for some $\varepsilon > 0$ there does not exist a finite set of points $x_1, x_2, x_3, \ldots, x_p$ so that

$$K \subset \bigcup_{i=1}^{p} B(x_i,\varepsilon).$$

(This follows from Exercise 13.7.1.) Thus we can choose a sequence of points x_1, x_2, x_3, ... in K with the property that

$$x_n \notin \bigcup_{i=1}^{n-1} B(x_i, \varepsilon).$$

Such a sequence can have no Cauchy subsequence since

$$d(x_N, x_m) > \varepsilon$$

for all $n > M$. Thus it has no convergent subsequence. Since this contradicts the Bolzano-Weierstrass property, the set K must be separable. ■

Note. In our proof we encountered a technical idea that is of independent interest. If K is compact, then we have shown that it is bounded in a strong sense: Given any radius ε we can cover K by a finite number of balls of that radius. More precisely, for every $\varepsilon > 0$ there exists $n \in \mathbb{N}$ and open balls

$$B(x_1, \varepsilon), \ldots, B(x_n, \varepsilon)$$

such that

$$K \subset \bigcup_{i=1}^{n} B(x_i, \varepsilon).$$

The set $\{x_1, x_2, \ldots, x_n\}$ is called an ε-*net* for K. It has the property that if $x \in K$ there exists i such that $d(x_i, x) < \varepsilon$. Clearly, this is a stronger condition than boundedness: Not merely is K contained in some large ball (i.e., is bounded) but K is contained in the union of a finite number of balls of any specified diameter. This will play a role in Section 13.12.4.

We might have hoped that the list of properties in Theorem 13.90, which are evidently necessary properties of compact sets, might also be sufficient. Theorem 13.89 gives a clue: Although compactness is a topological property, boundedness is not. A simple example also illustrates. Consider the real line $\mathbb{R}$ but furnished with a different metric from the usual, the metric of Exercise 13.2.1(c) or (d). Since that metric is equivalent to the usual metric, every closed subset of $\mathbb{R}$ has all the properties of Theorem 13.90; the only difference here is that under the new metric all sets are bounded. But the only closed subsets of $\mathbb{R}$ that are compact (under either metric) are the ones that are bounded in the usual metric.

Exercises

13.12.1 Show that every closed subset of a compact set is also compact.

13.12.2 Show that compactness is a topological property.
SEE NOTE 356

13.12.3 Show that every finite subset of a metric space is compact.

13.12.4 What subsets of the space (X, d), where d is the discrete metric, are compact?
SEE NOTE 357

13.12.5 Let $\{x_n\}$ be a convergent sequence in a metric space with limit z. Show that the set

$$\{z, x_1, x_2, x_3, \dots\}$$

is compact.

13.12.6 Prove that a subset of $\mathbb{R}^n$ is compact if and only if it is closed and bounded.

13.12.7 Show that the Hilbert cube is a compact subset of ℓ_2.

SEE NOTE 358

13.12.8 Show that the unit sphere in $\mathcal{C}[a,b]$, that is, the set

$$\{f \in \mathcal{C}[a,b] : |f(x)| \leq 1, \ \ x \in [a,b]\}$$

is not compact.

13.12.9 Show that if K is a compact subset of a metric space (X,d), then for any $x \in X$ there is a point $k \in K$ so that

$$d(k,x) = \inf\{d(x,y) : y \in K\}.$$

Show that if K is not compact, but merely closed, this would not necessarily be true. If K is complete but not compact, is this always true?

SEE NOTE 359

13.12.10 Show that if E and F are closed subsets of a metric space (X,d), at least one of which is compact, then there are points $e \in E$ and $f \in F$ so that

$$d(e,f) = \inf\{d(x,y) : x \in E, \ y \in F\}.$$

Show that if the sets E and F are not compact, but merely closed or complete, then this would not necessarily be true.

13.12.11 Show that the product of two compact metric spaces furnished with the product metric (Exercise 13.2.9) is also a compact metric space.

13.12.2 Continuous Functions on Compact Sets

Many of the standard theorems about continuous functions on compact subsets of $\mathbb{R}$ or $\mathbb{R}^n$ carry over to general metric spaces. If $f : [a,b] \to \mathbb{R}$ is continuous, then

1. f is uniformly continuous on $[a,b]$.
2. f is bounded.
3. f attains its maximum and minimum.

We ask now, what are the analogues of this for a continuous function

$$f : X \to Y$$

where X is a compact metric space?

Definition 13.91: If $f : (X,d) \to (Y,e)$ and for every $\varepsilon > 0$ there exists $\delta > 0$ such that $e(f(x), f(x')) < \varepsilon$ whenever $d(x,x') < \delta$, we say f is *uniformly continuous* on X.

Let us treat uniform continuity first. We prove, as for continuous functions defined on a compact subset of $\mathbb{R}$, that continuous functions on compact spaces are uniformly continuous. Since the proofs do not require new methods, they have been left as Exercises 13.12.17 and 13.12.18.

Theorem 13.92: *If X is compact and $f:X\to Y$ is continuous, then f is uniformly continuous.*

We now generalize the elementary theorem that asserts that a continuous real-valued function on a compact interval $[a,b]$ achieves absolute extrema on $[a,b]$. We can anticipate what form this will assume in a general metric space if we remember that for a continuous function $f:[a,b]\to\mathbb{R}$ the image set $f([a,b])$ must be itself a compact interval in $\mathbb{R}$.

Theorem 13.93: *If $f:X\to Y$ is continuous and X is compact, then the set $f(X)$ is compact in Y.*

Exercises

13.12.12 If $f:(X,d)\to(Y,e)$ and d is the discrete metric is f continuous? Is f uniformly continuous?

13.12.13 If $f:(X,d)\to(Y,e)$ and e is the discrete metric is f continuous? Is f uniformly continuous?

13.12.14 Show that every contraction mapping on a metric space is uniformly continuous.

13.12.15 Let (X,d) be a metric space and let A be a nonempty subset of X. Define $f:X\to\mathbb{R}$ by

$$f(x)=\operatorname{dist}(x,A)=\inf\{d(x,y):y\in A\}.$$

In Exercise 13.6.11 we established that f is continuous. Is f uniformly continuous?

13.12.16 Show that if $f:X\to Y$ is uniformly continuous and $\{x_n\}$ is a Cauchy sequence in X, then $\{f(x_n)\}$ is a Cauchy sequence in Y. Show that this need not be true if f is merely continuous.

13.12.17 Prove that if X is compact and $f:X\to Y$ is continuous, then f is uniformly continuous.

13.12.18 Prove that if $f:X\to Y$ is continuous and X is compact, then the set $f(X)$ is compact in Y.

13.12.19 Let X and Y be metric spaces with X compact. Prove that a continuous, one-to-one mapping of X onto Y is necessarily a homeomorphism.

13.12.20 Let $f:X\to X$ be a continuous mapping from a compact space X into itself. Define the sequence of sets

$$X_1=f(X),\ X_2=f(X_1),\ \ldots,\ X_n=f(X_{n-1}).$$

Let $K=\bigcap_{i=1}^{\infty}X_i$. Show that K is nonempty, compact, and invariant under f in the sense that $f(K)=K$.

13.12.21 A function $f : X \to X$ on a metric space (X,d) is a *weak contraction* if

$$d(f(x),f(y)) < d(x,y) \quad x \neq y.$$

(a) Show that such a function f must be uniformly continuous.

(b) Show that such a function f need not be a contraction.

(c) Show that if X is compact, then a weak contraction f must have a unique fixed point.

(d) Show that if X is complete but not compact, then a weak contraction f need not have a fixed point, but if it does that fixed point is unique.

(e) How does the result in (c) compare to the Banach fixed point theorem?

See Note 360

13.12.22 Formulate and prove a generalization of Dini's theorem (Theorem 9.24) for sequences of continuous real-valued functions on a compact metric space.

13.12.3 The Heine-Borel Property

✂ *Enrichment section. May be omitted.*

We recall that on the real line a compact set can be characterized in several different ways. One of the most useful for the purposes of compactness arguments was the Heine-Borel property, that any covering of a closed and bounded set by a family of open intervals can be reduced to a finite subcovering.

This too is available in a general metric space. It is no longer true for all closed and bounded sets, but it is true for all compact sets as we have defined them in the preceding sections.

Let X be a metric space, and let $K \subset X$. A collection $\mathcal{U}$ of open sets is called an *open cover* of K if

$$K \subset \bigcup_{U \in \mathcal{U}} U.$$

Theorem 13.94: *The following conditions on a set K in a metric space X are equivalent.*

1. **(Bolzano-Weierstrass Property)** *K is compact.*
2. **(Heine-Borel Property)** *Every open cover of K can be reduced to a finite subcover.*

Proof. Throughout we can assume that $X = K$. Let K satisfy (2), and let $\{x_n\}$ be a sequence in K. For each $N \in \mathbb{N}$, let $A_N = \{x_n : n \geq N\}$ and let $U_N = X \setminus \overline{A}_n$. Note that $\{U_N\}$ forms an increasing sequence of open sets and that $U_N \neq K$ for any N. In particular, each of the sets U_N is open and no finite collection of the sets U_N covers K. Since K satisfies condition (2), this cannot be an open cover of K and so $\bigcup_{N=1}^{\infty} U_N \neq X$; that is,

$$\bigcap_{N=1}^{\infty} \overline{A}_N \neq \emptyset.$$

Let $x_0 \in \bigcap_{N=1}^{\infty} \overline{A}_N$. It follows directly from the definition of the sets $\overline{A}_N$ that x_0 is the limit of some subsequence of the sequence $\{x_n\}$. This completes the proof of $(2) \Rightarrow (1)$.

Now suppose that K satisfies condition (1). We need to recall from Theorem 13.90 that K is separable as this is needed in order to prove that K has the Heine-Borel property. Now let $\mathcal{U}$ be an open cover of X. It follows from Lindelöf's theorem (Exercise 13.12.30) that $\mathcal{U}$ can be reduced to a countable subcover $\{U_1, U_2, U_3, \dots\}$.

We now show that this subcover can be further reduced to a finite subcover. If this were not the case, then for each $N \in \mathbb{N}$ there exists

$$x_N \in X \setminus \bigcup_{i=1}^{N} U_i$$

since otherwise the collection $\{U_1, U_2, U_3, \dots, U_N\}$ would cover X. Since X has the Bolzano-Weierstrass property, the set $\{x_1, x_2, \dots\}$ has a limit point x_0. But $X = \bigcup_{i=1}^{\infty} U_i$, so there exists $j \in \mathbb{N}$ such that $x_0 \in U_j$. This implies that $x_i \in U_j$ for infinitely many $i \in \mathbb{N}$. This is impossible because our choice of the points x_N implies that $x_N \in X \setminus U_j$ when $N \geq j$. This contradiction implies that the collection $U_1, U_2, \dots$ can be reduced to a finite subcover, completing the proof of $(1) \Rightarrow (2)$. ■

Exercises

13.12.23 Show that the following subsets of $\mathbb{R}$ do not have the Heine-Borel property by constructing an open cover with no finite subcover.

(a) the set $\mathbb{N}$

(b) the set $(0, 1)$

(c) the set $\mathbb{Q} \cap [0, 1]$

13.12.24 Without appealing to Theorem 13.94, prove directly that a set with the Heine-Borel property is bounded.

13.12.25 Without appealing to Theorem 13.94, prove directly that a set with the Heine-Borel property is closed.

13.12.26 Without appealing to Theorem 13.94, prove directly that a set with the Heine-Borel property is separable.

13.12.27 Use the Heine-Borel property to describe completely what sets are compact in a discrete metric space.

13.12.28 Use the Heine-Borel property to prove that if X is compact and $f : X \to Y$ is continuous, then f is uniformly continuous.

13.12.29 Use the Heine-Borel property to prove that if $f : X \to Y$ is continuous and X is compact, then the set $f(X)$ is compact in Y.

13.12.30 **(Lindelöf's Theorem)** Prove that every open cover of a separable metric space has a countable subcover.

SEE NOTE 361

13.12.31 Show that to every open cover $\mathcal{C}$ of a compact set K in a metric space there corresponds a positive number L (called the *Lebesgue number* of the cover) such that if $x, y \in K$ and $d(x,y) < L$, then there is some member $U \in \mathcal{C}$ such that both x and y belong to U.

13.12.32 Use Exercise 13.12.31 to prove that if X is compact and $f : X \to Y$ is continuous, then f is uniformly continuous.

13.12.33 Show that the property of Exercise 13.12.31 is not equivalent to compactness in $\mathbb{R}$ (i.e., find a noncompact set $K \subset \mathbb{R}$ with this property).

SEE NOTE 362

13.12.34 Show that a metric space (X,d) is compact if and only if for every family $\mathcal{F}$ of closed subsets of X for which

$$\bigcap_{F \in \mathcal{F}} F = \emptyset$$

there must be a finite collection $F_1, F_2, \ldots, F_m$ of sets in $\mathcal{F}$ so that

$$\bigcap_{i=1}^{m} F_i = \emptyset.$$

SEE NOTE 363

13.12.4 Total Boundedness

✂ *Enrichment section. May be omitted.*

In $\mathbb{R}^n$ a set is compact if and only if it is both closed and bounded. We have seen already that this is false in a general metric space. Rather than simply abandon this idea we still can pursue the notion that for a set to be compact it should be sufficient that it is closed and "bounded" in some stronger sense.

The key idea has already appeared in the proof of Theorem 13.90. There we showed that if X is compact, then, for every $\varepsilon > 0$, there is an ε-net, that is, a finite set

$$\{x_1, x_2, \ldots, x_n\} \subset X$$

such that the finite collection of balls $\{B(x_i, \varepsilon)\}$ covers X. When a space X has, for every $\varepsilon > 0$, an ε-net, we say that X is *totally bounded.* We express this formally.

Definition 13.95: Let X be a metric space. We say that a set $S \subset X$ is *totally bounded* if for every $\varepsilon > 0$ there is a finite set

$$\{x_1, x_2, \ldots, x_n\} \subset X$$

such that

$$S \subset B(x_1, \varepsilon) \cup B(x_2, \varepsilon) \cup \cdots \cup B(x_n, \varepsilon).$$

We can also characterize total boundedness in terms of Cauchy sequences. Note that the language of this theorem is close to the language of the Bolzano-Weierstrass property.

Theorem 13.96: *A metric space X is totally bounded if and only if every sequence has a Cauchy subsequence.*

Proof. Suppose that X is not totally bounded. Then for some $\varepsilon > 0$ there does not exist a finite set of points $x_1, x_2, x_3, \ldots, x_p$ so that

$$K \subset \bigcup_{i=1}^{p} B(x_i, \varepsilon).$$

Thus we can choose a sequence of points $x_1, x_2, x_3, \ldots$ in K with the property that

$$x_n \notin \bigcup_{i=1}^{n-1} B(x_i, \varepsilon).$$

Such a sequence can have no Cauchy subsequence since

$$d(x_N, x_m) > \varepsilon$$

for all $n > M$.

Conversely, if X is totally bounded, and $\{x_n\}$ is an arbitrary sequence, then we can construct a Cauchy subsequence as follows. X can be expressed as the union of a finite number of balls of radius 1. Our sequence $\{x_n\}$ must have a subsequence that is in one of these. That ball, in turn, can be subdivided using a finite number of balls of radius $1/2$. By choosing an appropriate subsequence of that subsequence we can arrive at a subsequence that is in a ball of diameter $1/2$. By continuing this process indefinitely and taking a sequence that is a subsequence of *all* of these we arrive at a final subsequence. We ask you, in Exercise 13.12.43, to give a precise description of this process and verify that the sequence constructed is a Cauchy subsequence of $\{x_n\}$. ■

We are now in a position to reinterpret our standard result that compact in $\mathbb{R}^n$ is equivalent to closed and bounded. Closed subsets of $\mathbb{R}^n$ are complete; bounded subsets of $\mathbb{R}^n$ are totally bounded. Thus we could equally say that compact subsets of $\mathbb{R}^n$ are those that are complete and totally bounded. Expressed this way we have a theorem that works in every metric space.

Theorem 13.97: *A metric space is compact if and only if it is complete and totally bounded.*

Proof. Suppose that X is compact. Let $\{x_n\}$ be a Cauchy sequence in X. By the Bolzano-Weierstrass property $\{x_n\}$ has a convergent subsequence. But a Cauchy sequence with a convergent subsequence is itself convergent (Exercise 13.8.6). Thus X is complete.

In a similar way we see that X is totally bounded: If $\{x_n\}$ is an arbitrary sequence in X, then it has a convergent subsequence. That subsequence is Cauchy, and so it follows immediately from Theorem 13.94 that X is totally bounded.

Conversely, suppose that X is complete and totally bounded. If $\{x_n\}$ is an arbitrary sequence from X, then $\{x_n\}$ has a Cauchy subsequence, by Theorem 13.96. This subsequence converges, since X is complete. Thus X is compact by definition. ■

Exercises

13.12.35 Show that a set $S \subset X$ is *totally bounded* if and only if for every $\varepsilon > 0$ there is a finite set $\{x_1, x_2, \ldots, x_n\} \subset S$ such that

$$S \subset B(x_1, \varepsilon) \cup B(x_2, \varepsilon) \cup \cdots \cup B(x_n, \varepsilon).$$

See Note 364

13.12.36 The mathematician Herman Weyl is credited with joking that a "compact city is a city that can be guarded by a finite number of arbitrarily near-sighted policemen." Explain.

13.12.37 Show that every subset of a totally bounded set is totally bounded.

13.12.38 Show that the closure of a totally bounded set is totally bounded.

13.12.39 What sets are totally bounded in a discrete metric space?

13.12.40 Show that every totally bounded set is bounded but that the converse is not true.

13.12.41 Show directly that a set in $\mathbb{R}^n$ is totally bounded if and only if it is bounded.

See Note 365

13.12.42 Show that total boundedness is not invariant under continuous mappings. Is it a topological property? Is it invariant under uniformly continuous mappings? Is it invariant under isometries?

13.12.43 Supply all details needed to complete the proof of Theorem 13.96.

See Note 366

13.12.44 Show that closed balls in $\mathcal{C}[a,b]$, $M[a,b]$, and ℓ_∞ are not compact by using Theorem 13.97.

13.12.45 Prove that a totally bounded metric space must be separable.

See Note 367

13.12.46 Compare the following properties that a metric space might have.

(a) Every closed and bounded subset is compact.

(b) Every closed and bounded subset is totally bounded.

(c) Every bounded sequence has a convergent subsequence.

(d) Every bounded sequence has a Cauchy subsequence.

See Note 368

13.12.47 Show that a closed and bounded set E in ℓ_2 is compact if and only if

$$\lim_{N\to\infty} \sum_{i=N}^{\infty} x_i^2 = 0$$

uniformly for $x = (x_1, x_2, x_3, \ldots) \in E$. State and prove the analogous version in the metric space ℓ_1.

13.12.5 Compact Sets in $\mathcal{C}[a,b]$

✂ *Enrichment section. May be omitted.*

In any metric space that is important to us it will be equally important to determine which sets are compact. Many theorems of existence and uniqueness in analysis can be obtained best from a compactness argument. Thus we shall need to know which sets in our space allow such arguments.

The space $\mathcal{C}[a,b]$ of continuous functions on an interval $[a,b]$ furnished with the supremum metric

$$d(f,g) = \max_{x\in X} |f(x) - g(x)|$$

is a complete separable metric space. Which subsets are compact?

While all compact sets must be closed and bounded, it is not true in this space that all closed and bounded sets are compact. Our purpose here is to obtain a useful characterization of the compact subsets of $\mathcal{C}[a,b]$. This characterization involves two properties that a family of functions on $[a,b]$ may or may not possess. For the first property, let us ask what characterizes the bounded subsets of $\mathcal{C}[a,b]$ since every compact set must also be bounded.

Definition 13.98: A family $\mathcal{F}$ of functions on an interval $[a,b]$ is said to be *uniformly bounded* on $[a,b]$ if there exists $M > 0$ such that $|f(x)| \leq M$ for all $x \in [a,b]$ and $f \in \mathcal{F}$.

This concept characterizes the bounded subsets of our metric space $\mathcal{C}[a,b]$. The proof is straightforward and is left to the exercises.

Lemma 13.99: *Let $\mathcal{F}$ be a family of continuous functions on $[a,b]$. Then $\mathcal{F}$ is uniformly bounded on $[a,b]$ if and only if $\mathcal{F}$ is a bounded subset of the metric space $\mathcal{C}[a,b]$.*

The other relevant notion concerns the uniformity of the continuity behavior of continuous functions in a compact subset of $\mathcal{C}[a,b]$. Let $f \in \mathcal{C}[a,b]$, let $x_0 \in X$, and let $\varepsilon > 0$. Then there exists $\delta > 0$ such that, if $|x - x_0| < \delta$, $|f(x) - f(x_0)| < \varepsilon$. The number δ depends on x_0, ε, and f and should perhaps be written $\delta = \delta(x_0, \varepsilon, f)$. Since any continuous function on $[a,b]$ is also uniformly continuous (Theorem 5.48), we know that δ can be chosen so as to be independent of x_0 for a given ε and f. If $\mathcal{F} \subset \mathcal{C}[a,b]$ and we can choose δ so as also to be independent of $f \in \mathcal{F}$, we say that $\mathcal{F}$ is an *equicontinuous* family. The concept is due to Giulio Ascoli (1843–1896).

Definition 13.100: A family $\mathcal{F}$ of functions on a metric space (X,d) is *equicontinuous* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that, if $x, y \in X$ and $d(x,y) < \delta$, then $|f(x) - f(y)| < \varepsilon$ for all $f \in \mathcal{F}$.

For an easy example, note that a collection of functions that satisfies a uniform Lipschitz condition is equicontinuous.

Example 13.101: Let $M > 0$ and let

$$\mathcal{F} = \{f : [a,b] \to \mathbb{R} : |f(x) - f(y)| \leq M|x - y| \text{ for all } x, y \in [a,b]\}.$$

Then $\mathcal{F}$ is an equicontinuous family of functions on $[a,b]$. It suffices to take $\delta = \varepsilon/M$. ◄

Our main theorem, usually attributed to both Ascoli and Cesare Arzelà (1847–1912), now uses the two concepts of uniform boundedness and equicontinuity to obtain a characterization of compactness in the space $\mathcal{C}[a,b]$.

Theorem 13.102 (Arzelà-Ascoli) *Let K be a closed subset of the metric space $\mathcal{C}[a,b]$. Then K is compact if and only if K is uniformly bounded and equicontinuous.*

Proof. Note first that, since K is assumed to be a closed subset of a space that we already know is complete, K itself is complete. By Theorem 13.97 K is compact if and only if it is totally bounded. Thus we need prove only that K is uniformly bounded and equicontinuous if and only if K is totally bounded.

Suppose first that K is totally bounded in $\mathcal{C}[a,b]$. Then K is bounded in $\mathcal{C}[a,b]$ and is therefore a uniformly bounded family of functions. We show that K is equicontinuous. Let $\varepsilon > 0$, and let $f_1, f_2, \ldots, f_n$ be an $(\varepsilon/3)$-net in K. Let $f \in K$. There exists $j \leq n$ such that

$$\max_{z\in[a,b]} |f(z) - f_j(z)| < \tfrac{1}{3}\varepsilon. \tag{28}$$

Then, for $x, y \in [a,b]$,

$$|f(x) - f(y)| \leq |f(x) - f_j(x)| + |f_j(x) - f_j(y)| + |f_j(y) - f(y)|. \tag{29}$$

Each of the functions f_i is uniformly continuous. Thus, since there are only finitely many of them, there exists $\delta > 0$ such that

$$|x-y| < \delta,\ 1 \leq i \leq n \Rightarrow\ |f_i(x) - f_i(y)| < \varepsilon/3. \tag{30}$$

It now follows from (28), (29), and (30) that $|f(x) - f(y)| < \varepsilon$ for all x, $y \in [a,b]$ with $|x-y| < \delta$ and all $f \in K$. This shows that K is equicontinuous.

To prove the converse, suppose that K is uniformly bounded and equicontinuous. We show that K is totally bounded. Let $\varepsilon > 0$. We shall find an ε-net $\mathcal{F}$ in the space for K. This means $\mathcal{F}$ must be a finite collection of functions in $\mathcal{C}[a,b]$ such that every member of K is closer to some member of $\mathcal{F}$ than $\varepsilon > 0$. (Note that we do not need to select the members of $\mathcal{F}$ from K although it is possible to do so.)

Choose $M \in \mathbb{N}$ such that $|g(x)| \leq M$ for all $x \in X$ and $g \in K$. Let $\varepsilon > 0$. Since K is equicontinuous, there exists $\delta > 0$ such that

$$|x-y| < \delta,\ g \in K\ \Rightarrow\ |g(x) - g(y)| < \varepsilon/4. \tag{31}$$

Using this number δ, we subdivide $[a,b]$ by points $x_1, x_2, \ldots, x_n$ sufficiently close together so that every point in $[a,b]$ is within δ of one of these points. (Note that this would be a δ-net for the compact metric space $[a,b]$.)

Choose $m \in \mathbb{N}$ such that $1/m < \varepsilon/4$, and partition the interval $[-M,M]$ into $2Mm$ congruent intervals:

$$-M = y_0 < y_1 < \cdots < y_{2Mm} = M.$$

Consider now the grid of points in the rectangle $[a,b] \times [-M,M]$ whose first coordinates are from the points $x_1, x_2, \ldots, x_n$ and whose second coordinates are from the

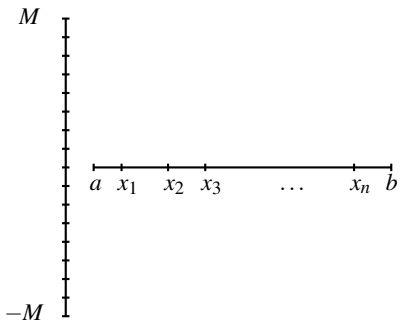


Figure 13.7. Partitioning the intervals $[-M,M]$ and $[a,b]$ in the proof of Theorem 13.102.

points $y_0, y_1, \ldots, y_{2Mm}$. Figure 13.7 illustrates the situation. Let $\mathcal{F}$ denote the set of all continuous functions on $[a,b]$ that are piecewise linear and whose corner points on the graph occur at points in the grid. There are clearly only finitely many such functions.

Let $g \in K$. There exists, because of the nature of the grid, at least one function $f \in \mathcal{F}$ so that

$$|g(x_j) - f(x_j)| < \varepsilon/4 \quad (j = 1,2,\ldots,n). \tag{32}$$

Consider any point $x \in [a,b]$. There is some interval with $x \in [x_j, x_{j+1}]$ for some $1 \leq j < n$. We know that

$$|g(x_j) - g(x_{j+1})| < \varepsilon/4$$

because of (31). It follows also then, because of (32), that

$$|f(x_j) - f(x_{j+1})| < \varepsilon/2.$$

As f is linear on the interval $[x_j, x_{j+1}]$ we have as well the inequality

$$|f(x_j) - f(x)| < \varepsilon/2$$

for all $x \in [x_j, x_{j+1}]$. Together these inequalities show that

$$|g(x) - f(x)| \leq |g(x) - g(x_j)| + |g(x_j) - f(x_j)| + |f(x_j) - f(x)| < \varepsilon$$

which implies that

$$\max_{x \in [a,b]} |f(x) - g(x)| < \varepsilon.$$

We have shown that $\mathcal{F}$ is an ε-net, so K is totally bounded, as was to be proved. ■

Exercises

13.12.48 Prove that a family $\mathcal{F}$ of continuous functions on an interval $[a,b]$ is uniformly bounded if and only if $\mathcal{F}$ is a bounded subset of the metric space $\mathcal{C}[a,b]$.

13.12.49 Is the set of functions $\{\sin kt : k = 1,2,3,\ldots\}$ equicontinuous on $[0,2\pi]$?
SEE NOTE 369

13.12.50 Let $f:\mathbb{R}\to\mathbb{R}$ be uniformly continuous on $\mathbb{R}$ and let f_a denote the function $f_a(t)=f(t-a)$. Show that the family $\{f_a : a\in\mathbb{R}\}$ is equicontinuous on $\mathbb{R}$. Is this true if f is merely continuous?

13.12.51 Prove directly from the definition that the family of functions $f_n(x)=x^n$, $n=1,2,3,\ldots$ is not equicontinuous on the interval $[0,1]$ but is equicontinuous on the interval $[0,1/2]$.

13.12.52 Let A be a bounded subset of $\mathcal{C}[a,b]$. Prove that the set of all functions of the form

$$F(x)=\int_a^x f(t)\,dt$$

for $f\in A$ is an equicontinuous family.

13.12.53 Let K be continuous on $[a,b]\times[a,b]$ and ϕ continuous on $[a,b]$. The operator $A:\mathcal{C}[a,b]\to\mathcal{C}[a,b]$ is defined by

$$(A(f))(x)=\lambda\int_a^b K(x,y)f(y)\,dy+\phi(x)$$

(cf. Example 13.81). Show that the image under A of any bounded set in $\mathcal{C}[a,b]$ is an equicontinuous family.

13.12.54 Let σ be continuous and nondecreasing on $[0,\infty)$, with $\sigma(0)=0$. A function $f\in\mathcal{C}[a,b]$ has *modulus of continuity* σ if

$$|f(x)-f(y)|\le\sigma(|x-y|)$$

for all $x,y\in[a,b]$. Let $C(\sigma)$ denote

$$\{f : \sigma \text{ is a modulus of continuity for } f\}.$$

(a) Show that every $f\in\mathcal{C}[a,b]$ has a modulus of continuity.

(b) Let σ be a modulus of continuity. Show that $C(\sigma)$ is an equicontinuous family.

(c) Exhibit a modulus of continuity for the class of Lipschitz functions with constant M.

(d) Let σ be a modulus of continuity. Is it necessarily true that $\sigma\in C(\sigma)$ on $[a,b]$? What if σ is concave down?

(e) Prove that the set

$$K=\left\{f\in\mathcal{C}[0,1] : |f(x)-f(y)|\le\sqrt{|x-y|} \text{ and } f(0)=0\right\}$$

is a compact subset of $\mathcal{C}[0,1]$. Is $\sqrt{x}\in K$? What about x^2?

13.12.55 Let $Lip_1[a,b]$ denote the family of functions f on $[a,b]$ that satisfy the Lipschitz condition

$$|f(x)-f(y)|\le|x-y|$$

for all x, $y\in[a,b]$. We furnish this space with the usual supremum metric so that it is a subspace of $\mathcal{C}[a,b]$. Prove that a sequence of functions $\{f_n\}$ converges in $Lip_1[a,b]$ if and only if it converges pointwise.

See Note 370

13.12.56 Let $\{f_n\}$ be a sequence of real-valued functions defined on an interval $[a,b]$ and suppose that $\{f_n(x)\}$ is bounded for each $x\in[a,b]$.

(a) Show that there is a subsequence $\{f_{n_k}\}$ so that $\lim_{k\to\infty}f_{n_k}(r)$ exists for every rational number $r\in[a,b]$.

(b) Show that, if $\{f_n\}$ is equicontinuous, then in fact $\lim_{k\to\infty} f_{n_k}(x)$ exists for every $x \in [a,b]$ and the convergence is uniform.

(c) Use this to give another proof of the Arzelà-Ascoli theorem.

SEE NOTE 371

13.12.57 Use the Arzelà-Ascoli theorem to show that the set

$$\{f \in \mathcal{C}[0,1] : |f(x)| \leq 1\}$$

in $\mathcal{C}[0,1]$ cannot be covered by a sequence of compact sets. (This is also done, using the Baire category theorem, in Exercise 13.13.9.)

13.12.58 Prove the following more general version of the Arzelà-Ascoli theorem:

> *Let K be a closed subset of the metric space $C(X)$, where X is a compact metric space and $C(X)$ denotes the space of continuous real-valued functions on X furnished with the supremum metric. Then K is compact in $C(X)$ if and only if K is uniformly bounded and equicontinuous.*

SEE NOTE 372

13.12.6 Peano's Theorem

✂ *Enrichment section. May be omitted.*

In Section 13.11.4 we saw how the contraction mapping principle can be used to prove an existence and uniqueness theorem for solutions to the differential equation $y' = f(x,y)$. The requirement that was needed in order to apply the contraction mapping principle involved a Lipschitz condition on the function f.

In order to obtain an existence theorem (this time without uniqueness) under weaker hypotheses we need a different approach. For this we go back to a simple idea of Euler from 1768. To solve numerically an initial value problem

$$y' = f(x,y) \quad y(x_0) = y_0$$

on an interval $[x_0,b]$, divide the interval into n equally spaced points (equal subdivision is not an essential feature, but is traditional in numerical methods)

$$x_0 < x_1 < x_2 < \cdots < x_n = b$$

and approximate a solution by a piecewise linear function. Write

$$y_1 = y_0 + f(x_0,y_0)(x_1 - x_0),$$

$$y_2 = y_1 + f(x_1,y_1)(x_2 - x_1),$$

and so on through to

$$y_n = y_{n-1} + f(x_{n-1},y_{n-1})(x_n - x_{n-1}).$$

We can let $k_n(x)$ denote the function on $[x_0,b]$ that is continuous, passes through the points (x_0,y_0), (x_1,y_1),...,(x_n,y_n), and is linear in between. Note that $k_n'(x) = f(x_i,k(x_i))$ if x is in the ith interval (x_i,x_{i+1}) and that there is no derivative at the corner points.

In practice this method gives a reasonable approximation to solutions. We could make this the basis of an existence proof if we could show that the sequence $\{k_n\}$ converges uniformly to a function k and that the function k solves the initial value problem.

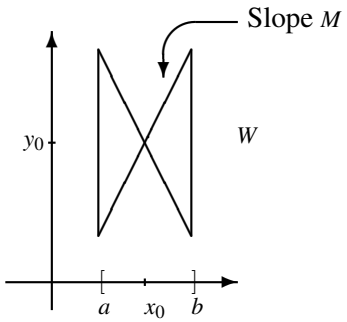


Figure 13.8. The set W and its projection to $[a,b]$ in the proof of Theorem 13.103.

Unfortunately, the hypothesis that we wish to use, the continuity of the function f, is too weak to allow a proof that the sequence $\{k_n\}$ converges uniformly. But we can use a compactness argument to obtain a uniformly convergent subsequence. The key is to use the continuity of f to design an interval $[x_0,b]$ on which such a sequence of functions $\{k_n\}$ is bounded and equicontinuous. Then the Arzelà-Ascoli theorem supplies the subsequence.

This improvement of the classical existence theorems of Cauchy and Lipschitz was obtained by Peano in 1886. In 1890 he showed that this weakening of hypotheses occurs at the expense of uniqueness by supplying the example that we consider in Exercise 13.12.59.

Theorem 13.103 (Peano) *Let f be continuous on an open subset D of $\mathbb{R}^2$, and let $(x_0,y_0) \in D$. Then the differential equation*

$$y' = f(x,y)$$

has a local solution passing through the point (x_0,y_0).

Proof. We are seeking an exact solution that is valid in some interval containing the point x_0. Thus we wish to find an interval $[a,b]$ containing x_0 and a differentiable function k defined on $[a,b]$ such that

$$k(x_0) = y_0 \text{ and } k'(x) = f(x,k(x)) \text{ for all } x \in [a,b]. \tag{33}$$

Our strategy is essentially to construct a family K of approximate solutions through (x_0,y_0) on $[a,b]$. We then show that the set $\overline{K}$ is compact in $C([a,b])$ and use compactness to show the existence of the function k as a point in $\overline{K}$.

We first select the interval $[a,b]$. Let R be a closed rectangle contained in D having sides parallel to the coordinate axes and having (x_0,y_0) as center. Let $M \geq 1$ be an upper bound for $|f|$ on R. Let

$$W = \{(x,y) \in R : |y-y_0| \leq M|x-x_0|\},$$

and let $[a,b]$ be the projection of W onto the x-axis, as in Figure 13.8.

We next obtain a family K of functions defined on $[x_0,b]$ that can be considered as approximate solutions to (33). Since W is compact in $\mathbb{R}^2$, f is uniformly continuous on

W. Thus, for every $\varepsilon > 0$, there exists $\delta \in (0,1)$ such that, if $(x,y) \in W$ and $(\bar{x},\bar{y}) \in W$ with $|x-\bar{x}| < \delta$ and $|y-\bar{y}| < \delta$, then

$$|f(\bar{x},\bar{y}) - f(x,y)| < \varepsilon.$$

Choose points $x_1, x_2, \ldots, x_n$ such that

$$x_0 < x_1 < x_2 < \cdots < x_n = b \text{ and } |x_i - x_{i-1}| < \delta/M$$

for all $i = 1, \ldots, n$. Define a function k_ε on $[x_0, b]$ as follows: $k_\varepsilon(x_0) = y_0$ and, on $[x_0, x_1]$, k_ε is linear with slope $f(x_0, y_0)$; on $[x_1, x_2]$, take k_ε to be linear with slope $f(x_1, k_\varepsilon(x_1))$; continuing in this way, we extend the definition of k_ε to all of $[x_0, b]$.

We have arrived at a function k_ε defined on $[x_0, b]$ whose graph is a polygonal arc through the point (x_0, y_0) and is contained in W. Since the slopes of the line segments composing the graph of k_ε are determined by values of the function f in W, we see that

$$|k_\varepsilon(x) - k_\varepsilon(\bar{x})| \leq M|x - \bar{x}| \tag{34}$$

for all $x, \bar{x} \in [x_0, b]$. Now let $x \in [x_0, b]$, $x \neq x_i$, $i = 0, 1, \ldots, n$. Then there exists $j \in \{1, 2, \ldots, n\}$ such that $x_{j-1} < x < x_j$. Noting that

$$|x_j - x_{j-1}| < \delta/M$$

and using (34), we see that

$$|k_\varepsilon(x) - k_\varepsilon(x_{j-1})| \leq M|x - x_{j-1}| < \delta.$$

This implies that

$$|f(x_{j-1}, k_\varepsilon(x_{j-1})) - f(x, k_\varepsilon(x))| < \varepsilon.$$

But

$$k'_\varepsilon(x) = f(x_{j-1}, k_\varepsilon(x_{j-1})),$$

so

$$|k'_\varepsilon(x) - f(x, k_\varepsilon(x))| < \varepsilon. \tag{35}$$

The inequality (35) is valid for all $x \in [x_0, b]$ except at points x in the finite set $\{x_0, \ldots, x_n\}$, at which k_ε need not be differentiable. By (35), we see that the functions k_ε are approximate solutions to (33).

We have constructed a family K of functions, one function corresponding to every $\varepsilon > 0$. The family K is uniformly bounded on $[x_0, b]$, since the graph of each of the functions k_ε is contained in W. It follows from (34) that K is an equicontinuous family. The Arzelà-Ascoli theorem now implies that $\overline{K}$ is compact in $\mathcal{C}[x_0, b]$.

We can now complete the proof of the theorem. For all $x \in [x_0, b]$, we have

$$\begin{aligned} k_\varepsilon(x) &= y_0 + \int_{x_0}^{x} k'_\varepsilon(t)\, dt \\ &= y_0 + \int_{x_0}^{x} (f(t, k_\varepsilon(t)) + (k'_\varepsilon(t) - f(t, k_\varepsilon(t)))\, dt. \end{aligned} \tag{36}$$

The fact that k'_ε may fail to exist on the set $\{x_0, x_1, \ldots, x_n\}$ does not affect the integral.

Since $K \subset \overline{K}$ and $\overline{K}$ is compact the sequence $\{k_{(1/n)}\}$ contains a subsequence $\{k_{1/n_i)}\}$ that converges uniformly to some function $k \in \overline{K}$. Note that k must be continuous on

$[x_0, b]$. Since f is uniformly continuous on W, the functions $f(t, k_{(1/n_i)}(t))$ converge uniformly to the function $f(t, k(t))$ on $[x_0, b]$. Noting (35), we now infer from (36) that

$$k(x) = y_0 + \int_{x_0}^{x} f(t, k(t))\, dt$$

for all $x \in [x_0, b]$. It follows that k is a solution to (33) on $[x_0, b]$.

In a similar manner, we obtain a solution $\overline{k}$ to (33) on $[a, x_0]$. The function y given by

$$y(x) = \begin{cases} k(x) & \text{for } x \in [x_0, b]; \\ \overline{k}(x) & \text{for } x \in [a, x_0], \end{cases}$$

satisfies (33) on all of $[a, b]$, as required. ■

Exercises

13.12.59 **(Peano's Example)** Show that the hypotheses given in Theorem 13.103 are not sufficient to guarantee uniqueness of solutions to the equation $y' = f(x, y)$ by taking, for example, the equation $y' = 3y^{2/3}$, $y(0) = 0$. Does this example conflict with the uniqueness assertion of Theorem 13.86?

SEE NOTE 373

13.13 Baire Category Theorem

In Section 6.4 we proved the Baire category theorem relative to the metric space $\mathbb{R}$. We are now in a position to establish this same theorem relative to any complete metric space. We introduce some of the basic ideas central to that theorem and prove the theorem in this section. Then, in Section 13.14, we provide some applications of this theorem. You may wish to review Sections 6.3 and 6.4 for a study of these ideas in the setting of the real line before proceeding to a development of the same ideas in an abstract metric space.

We begin by recalling the definition of a nowhere dense set in a metric space (as defined in Section 13.5 and used already in Exercises 13.5.21 and 13.6.21). After some examples of nowhere-dense sets we proceed to a proof of the Baire category theorem, which asserts that a complete space cannot be the union of any sequence of nowhere dense sets.

13.13.1 Nowhere Dense Sets

On the real line a set A is nowhere dense if it is dense in no interval; that would mean that, given any interval (a, b) there would be a subinterval $(c, d) \subset (a, b)$ that contains no points of A. In a metric space the role of the open intervals is played by open balls. Here is the formal definition.

Definition 13.104: Let (X, d) be a metric space. A set $A \subset X$ is called *nowhere dense* if, given any open ball $B(x, \varepsilon)$ in X, there exists an open ball $B(y, \delta) \subset B(x, \varepsilon)$ such that $A \cap B(y, \delta) = \emptyset$.

From this definition we see that A is nowhere dense provided that it fails to be dense in any open ball. It is easy to verify (Exercise 13.13.6) that A is nowhere dense if and only if $\overline{A}$ has empty interior. Thus a closed set is nowhere dense if and only if its complement is dense.

Let's look at some examples.

Example 13.105: Let K be the Cantor set in the interval $[0,1]$. Considered as a subset of $[0,1]$ it is clearly nowhere dense. [Be careful here: If instead the metric space is (K,d), where d is the usual metric, then K is not a nowhere dense set, it is dense.] ◄

Example 13.106: We shall show that the set $A = C[a,b]$ of all continuous functions taken as a subset of $M[a,b]$, the space of bounded functions, is nowhere dense. (Again notice the fact that we consider this in the space $M[a,b]$, not in $C[a,b]$.) Let $B(f,\varepsilon)$ be any open ball in $M[a,b]$. We shall find another open ball $B(g,\delta)$ contained in $B(f,\varepsilon)$ but containing no members of A, that is, this ball $B(g,\delta)$ must contain *no* continuous functions.

Now f, the center of $B(f,\varepsilon)$, may or may not be in A. We choose g as follows. If

$$\limsup_{t\to a} f(t) = \liminf_{t\to a} f(t), \tag{37}$$

then let

$$g(t) = \begin{cases} f(t)+\varepsilon/2, & \text{if } t \in [a,b]\cap\mathbb{Q} \\ f(t)-\varepsilon/2, & \text{if } t \in [a,b]\setminus\mathbb{Q}, \end{cases} \tag{38}$$

so

$$\limsup_{t\to a} g(t) - \liminf_{t\to a} g(t) = \varepsilon.$$

If, instead

$$\limsup_{t\to a} f(t) - \liminf_{t\to a} f(t) = \gamma > 0, \tag{39}$$

then let

$$g(t) = f(t) \text{ for all } t \in [a,b]. \tag{40}$$

We now obtain δ, for which $B(g,\delta)$ is the required ball. When (37) applies, take $\delta = \varepsilon/3$. When (39) applies, take $\delta = \gamma/3$. We must show

$$B(g,\delta) \subset B(f,\varepsilon) \tag{41}$$

and

$$B(g,\delta)\cap A = \emptyset. \tag{42}$$

To verify (41), we need only observe that

$$|g(t)-f(t)| = \frac{\varepsilon}{2} < \varepsilon$$

for all $t \in [a,b]$ when (38) applies and

$$|g(t)-f(t)| = 0 < \varepsilon$$

for all $t \in [a,b]$ when (40) applies.

To verify (42), let $h \in B(g,\delta)$. We show that $h \notin A$ by showing that

$$\limsup_{t\to a} h(t) - \liminf_{t\to a} h(t) > 0.$$

Now $h \in B(g,\delta)$, so $|g(t)-h(t)| < \delta$ for all $t \in [a,b]$. Thus

$$\limsup_{t\to a} h(t) \geq \limsup_{t\to a} g(t) - \delta \tag{43}$$

and

$$\liminf_{t\to a} h(t) \leq \liminf_{t\to a} g(t) + \delta. \tag{44}$$

Subtracting (44) from (43) we obtain

$$\limsup_{t\to a} h(t) - \liminf_{t\to a} h(t) \geq \limsup_{t\to a} g(t) - \liminf_{t\to a} g(t) - 2\delta.$$

When (38) applies, $2\delta = 2\varepsilon/3 < \varepsilon$, and when (40) applies, $2\delta = 2\gamma/3 < \gamma$. It now follows from (37) and (38) or (39) and (40) that

$$\limsup_{t\to a} h(t) - \liminf_{t\to a} h(t) > 0$$

so $h \notin A$. ◀

We observe that we could have presented a slightly simpler proof by showing that A is closed in $M[a,b]$ and that the complement of the set A is dense in $M[a,b]$. We leave this as Exercise 13.13.8. We provide such an argument in our next example.

Example 13.107: Recall the space ℓ_∞ of bounded sequences $\{x_i\}$ with metric

$$d_\infty(x,y) = \sup_i |x_i - y_i|.$$

(See Example 13.9.) We show that the subset c of convergent sequences is nowhere dense in ℓ_∞. To do this, we show that c is closed and its complement is dense.

To verify that c is closed, we show that its complement $\ell_\infty \setminus c$ is open. Let $x \in \ell_\infty \setminus c$. Then

$$L = \limsup_i x_i > \liminf_i x_i = l.$$

Take $\varepsilon = (L-l)/3$. If $d_\infty(x,y) < \varepsilon$, then

$$\limsup_i y_i \geq L - \frac{\varepsilon}{3} > l + \frac{\varepsilon}{3} \geq \liminf_i y_i,$$

so $y \in \ell_\infty \setminus c$. This proves that $\ell_\infty \setminus c$ is open: Each $x \in \ell_\infty \setminus c$ has a neighborhood contained in $\ell_\infty \setminus c$.

It remains to check that the complement of c is dense. Let $B(x,\varepsilon)$ be an open ball in ℓ_∞. We wish to show that $B(x,\varepsilon)$ contains points in $\ell_\infty \setminus c$. If $x \notin c$, there is nothing further to prove, so assume that $x \in c$. Let $L = \lim_i x_i$. Then there exists $N \in \mathbb{N}$ such that $|x_i - L| < \varepsilon/2$ if $i \geq N$. Consider the sequence $y = \{y_i\}$ such that

$$y_i = \begin{cases} x_i, & \text{if } i < N \\ L + \varepsilon/2, & \text{if } i \geq N,\ i \text{ odd} \\ L - \varepsilon/2, & \text{if } i \geq N,\ i \text{ even.} \end{cases}$$

Then $d_\infty(x,y) = \sup_i |x_i - y_i| < \varepsilon$, so $y \in B(x,\varepsilon)$. But

$$\limsup_i y_i = L + \frac{\varepsilon}{2} > L - \frac{\varepsilon}{2} = \liminf_i y_i,$$

so $y \notin c$. It is clear that $y \in \ell_\infty$. Thus $\ell_\infty \setminus c$ is dense in ℓ_∞.

We have shown c is closed and $\ell_\infty \setminus c$ is dense, from which it follows that c is nowhere dense in ℓ_∞. ◀

Exercises

13.13.1 Show that every subset of a nowhere dense set is also nowhere dense.

13.13.2 Show that the closure of every nowhere dense set is also nowhere dense.

13.13.3 Show that if $A_1, \ldots, A_n$ are nowhere dense subsets of a metric space X, then $A_1 \cup \cdots \cup A_n$ is also nowhere dense in X.

13.13.4 Is it true in an arbitrary metric space that every finite set is nowhere dense?
SEE NOTE 374

13.13.5 Describe what property a metric space must have in order that every finite set is nowhere dense.
SEE NOTE 375

13.13.6 (a) Show that a set A in a metric space X is nowhere dense if and only if $\overline{A}$ has empty interior.

(b) Show that a closed set in a metric space X is nowhere dense if and only if its complement is dense in X.

13.13.7 Show that the Hilbert cube (Exercise 13.4.4) is a nowhere-dense subset of ℓ_2.

13.13.8 Show that $\mathcal{C}[a,b]$ is nowhere dense in $M[a,b]$ by verifying that $\mathcal{C}[a,b]$ is closed and its complement is dense.

13.13.9 Use the Baire category theorem to show that the set

$$\{f \in \mathcal{C}[0,1] : |f(x)| \leq 1\}$$

in $\mathcal{C}[0,1]$ cannot be covered by a sequence of compact sets. (This is also done, using the Arzelà-Ascoli theorem, in Exercise 13.12.57.)

13.13.10 Show that the function $f : \ell_2 \to \ell_2$ defined on Hilbert space by specifying

$$f(x_1,x_2,x_3,x_4,\ldots) = (0,x_1,x_2,x_3,\ldots)$$

for every $(x_1,x_2,x_3,x_4,\ldots) \in \ell_2$ is an isometric map of ℓ_2 to a nowhere dense subset.

13.13.2 The Baire Category Theorem

On the real line the Baire category theorem asserts that any countable union of nowhere dense sets is small, so small in fact that its complement is dense. We might expect that this generalizes to a general metric space. All that needs to be added is the hypothesis that the space is complete.

Thus we have the statement of the Baire category theorem in a general metric space.

Theorem 13.108 (Baire) *Let (X,d) be a complete metric space, and let S be a countable union of nowhere dense sets in X. Then $X \setminus S$ is dense in X.*

Proof. Let $S = \bigcup_{n=1}^{\infty} S_n$, where each of the sets S_n is nowhere dense, and let B_0 be a nonempty open ball in X. To show that $X \setminus S$ is dense in X we must prove that

$$(X \setminus S) \cap B_0 \neq \emptyset.$$

Choose, inductively, a nested sequence of balls

$$B_n = B_n(x_n, r_n)$$

with $r_n < 1/n$ such that

$$\overline{B}_{n+1} \subset B_n \setminus \overline{S}_{n+1}.$$

To see that this is possible, first note that

$$B_n \setminus \overline{S}_{n+1} \neq \emptyset,$$

since S_{n+1}, and therefore $\overline{S}_{n+1}$, is nowhere dense. Thus we can choose

$$x_{n+1} \in B_n \setminus \overline{S}_{n+1}.$$

Since $\overline{S}_{n+1}$ is closed,

$$\operatorname{dist}(x_{n+1}, S_{n+1}) > 0,$$

so we can choose B_{n+1} as required. The sequence $\{x_n\}$ is a Cauchy sequence since, for $n, m \geq N$,

$$d(x_n, x_m) \leq d(x_n, x_N) + d(x_N, x_m) < 2/N.$$

Because X is complete, there exists $x \in X$ such that $x_n \to x$. But

$$x_{n+1} \in \overline{B}_{n+1}$$

for all n, so

$$x \in \bigcap_{n=1}^{\infty} \overline{B}_n \subset B_0 \cap (X \setminus S),$$

as was to be proved. ■

The following terminology is standard:

Definition 13.109: Let (X, d) be a metric space.

1. A set $A \subset X$ is called a *first category* set if A is a countable union of nowhere dense sets.

2. A set that is not of the first category is called a set of the *second category*.

3. The complement of a first-category set is called a *residual* set.

For complete metric spaces, first-category sets are the "small" sets and residual sets are the "large" sets in the sense of category. Second-category sets are merely "not small." Residual sets are large because they are dense and any intersection of a sequence of residual sets is still dense. First-category sets are small since (while they might be dense) their complement is always dense and any union of a sequence of first-category sets still has a dense complement.

For spaces that are not complete, a residual set can be empty. Note also that all these statements about sets in a space are relative to the space being considered. A set

might be first category as a subset of some space but not so as a subset of some other space.

Example 13.110: The set of rational numbers $\mathbb{Q}$ considered as a subset of $\mathbb{R}$ is dense. It is also first category since each singleton set $\{q\}$ for $q \in \mathbb{Q}$ is a nowhere dense subset of $\mathbb{R}$ and

$$\mathbb{Q} = \bigcup_{q \in \mathbb{Q}} \{q\} \tag{45}$$

expresses $\mathbb{Q}$ as a countable union of nowhere dense sets. Note that this also means that the set of irrational numbers is residual. ◀

Example 13.111: The entire incomplete metric space $\mathbb{Q}$ with the usual real metric is of the first category. Equation (45) again serves to display this space as a countable union of sets nowhere dense in $\mathbb{Q}$. Thus the complement of $\mathbb{Q}$ in $\mathbb{Q}$, namely the empty set $\emptyset$, would be by our definition a residual set. (In an incomplete metric space a residual set need not be "large.") ◀

Example 13.112: Consider the subspace $\mathbb{N}$ of $\mathbb{R}$. As a subset of $\mathbb{R}$, $\mathbb{N}$ is of the first category, since $\{n\}$ is nowhere dense in $\mathbb{R}$ for each n and

$$\mathbb{N} = \bigcup_{n=1}^{\infty} \{n\} \tag{46}$$

expresses $\mathbb{N}$ as a countable union of nowhere dense sets. But considered as a space in itself, $\mathbb{N}$ is a complete metric space. By the Baire category theorem it cannot be expressed as a countable union of nowhere dense sets. But how can that be? Doesn't equation (46) express $\mathbb{N}$ as a countable union of nowhere dense sets? In fact, note that in the space $\mathbb{N}$ each set $\{n\}$ is dense in $B(n, \frac{1}{2})$. The only residual set in $\mathbb{N}$ is $\mathbb{N}$ itself and only the empty set $\emptyset$ is nowhere dense or first category. ◀

Exercises

13.13.11 Show that every subset of a first category set also first category.

13.13.12 The closure of every nowhere dense set is also nowhere dense. Is the closure of every first category set also first category?

13.13.13 In a metric space (X,d), where d is the discrete metric, determine which sets are nowhere dense, first category, or residual.

13.13.14 In a metric space, show that a countable union of first category sets is again first category and that a countable intersection of residual sets is again residual.

13.13.15 In a metric space, show that every dense set of type $\mathcal{G}_\delta$ is residual.

13.13.16 Let $\mathcal{P}$ denote the polynomials on $[a,b]$, and let $\mathcal{P}_n \subset \mathcal{P}$ denote the polynomials of degree at most n. Show that $\mathcal{P}_n$ is nowhere dense in $\mathcal{C}[a,b]$. Deduce that $\mathcal{P}$ is a first category subset of $\mathcal{C}[a,b]$.

13.13.17 Show that a complete metric space with no isolated points must be uncountable.

13.13.18 Let c_0 denote the set of all sequences of real numbers converging to zero, that is,

$$c_0 = \{x \in c : \lim_{i\to\infty} x_i = 0\}.$$

Prove that c_0 is nowhere dense in c.

13.13.19 Let c_Q denote the set of all sequences of real numbers converging to a rational number, that is,

$$c_Q = \{x \in c : \lim_{i\to\infty} x_i \in \mathbb{Q}\}.$$

Is c_Q nowhere dense in c? Is c_Q of the first category in c?

13.14 Applications of the Baire Category Theorem

✂ *Enrichment section. May be omitted.*

The Baire category theorem has numerous applications. Category arguments taken together with compactness arguments are among the most powerful and commonly used tools in mathematical analysis.

We have already encountered some simple applications of the Baire category theorem (Theorem 6.13, Exercises 6.4.7 and 13.13.17). The first two are theorems that show that certain "pointwise" conditions actually hold "uniformly" on some open set. The third shows that a complete metric space without isolated points cannot be countable.

In this section we emphasize the use of the Baire category theorem to prove the existence of objects that might be difficult to imagine or to construct. The method is to view these objects as members of an appropriate complete metric space, and then to show the objects with the required property form a residual subset of that space.

In this we are reminded of a argument of Cantor's. Do transcendental real numbers exist? The nontranscendental numbers (i.e., the algebraic numbers) form a countable set of real numbers. What remains is a large dense set of numbers, so that, indeed, transcendental real numbers do exist and they exist in abundance. (See Exercise 2.3.10 for the details.) Here we apply essentially the same reasoning to a complete metric space: The objects whose existence we seek to prove are those objects in the space that remain after a first category set is removed. By the Baire category theorem they must exist in abundance, forming a dense and residual set in the space.

13.14.1 Functions Whose Graphs "Cross No Lines"

We begin with an example in which the objects are continuous functions.

The continuous functions we encounter in elementary calculus have the property that there are many straight lines that cross the graph of the function. Let us make the notion of "crossing lines" precise. We mean simply that the line is above the graph on one side of a point and below on the other. The formal definition looks rather formidable but simply expresses this elementary idea.

Definition 13.113: Let $f : [a,b] \to \mathbb{R}$, and let $L : \mathbb{R} \to \mathbb{R}$ be a function whose graph is a straight line. We say *L crosses f* or (*f crosses L*) if there exists $x_0 \in [a,b]$ and $\delta > 0$ such that $f(x_0) = L(x_0)$ and either

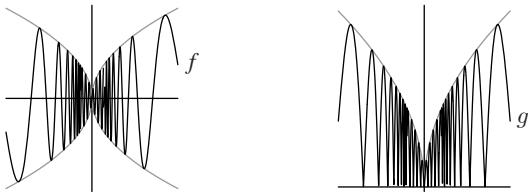


Figure 13.9. Graphs of $f(x) = \sqrt{|x|}\sin(1/x)$ and $g(x) = |f(x)|$. The asymptotes $y = \pm\sqrt{|x|}$ are also shown.

(i) $L(x) \le f(x)$ for all $x \in [x_0 - \delta, x_0] \cap [a,b]$ and
$L(x) \ge f(x)$ for all $x \in [x_0, x_o + \delta] \cap [a,b]$, or

(ii) $L(x) \ge f(x)$ for all $x \in [x_0 - \delta, x_0] \cap [a,b]$ and
$L(x) \le f(x)$ for all $x \in [x_0, x_o + \delta] \cap [a,b]$.

Example 13.114: Consider the function $f(x) = \sin x$. At the point $x = 0$ the x-axis (i.e., the horizontal line $y = 0$), crosses f since in the interval $(-\pi, \pi)$ the line is above the graph of f on the left on $(-\pi, 0)$ and below on the right on $(0, \pi)$. Indeed every line through $(0,0)$ crosses f. At the point $(\pi/2, 1)$ every line through that point except the horizontal line $y = 1$ crosses f. Since the line $y = 1$ stays always above the graph of f, it does not cross. ◀

The situation in the example is typical of the case for functions familiar in calculus. Every line crosses the graph of such functions, except perhaps the tangent line may not.

Example 13.115: It is easy to give examples of continuous functions that "wiggle" so much near a specific point that it is impossible for any line to cross the graph at that point. The functions

$$f(x) = \sqrt{|x|}\sin(1/x), \quad f(0) = 0$$

and $g(x) = |f(x)|$ illustrate this at the point $(0,0)$. Note that at every other point on the graph of f or g there are many lines that cross (Fig. 13.9). ◀

Although it may be difficult to imagine a continuous function that crosses *no* lines, and more difficult to construct one, we shall see that "most" (in the sense of category) continuous functions have this property. In Sections 13.14.2 and 13.14.3 we shall use this to show that "most" continuous functions fail to be monotonic on any interval and fail to be differentiable at any point.

Before stating Theorem 13.116, we introduce a bit of notation that will be convenient for our proof. Let $f \in C[a,b]$ and let $\gamma \in \mathbb{R}$. Define a function $f_{-\gamma}$ by

$$f_{-\gamma}(x) = f(x) - \gamma x.$$

Thus, $f_{-\gamma}$ is obtained from f by subtracting the linear function $L(x) = \gamma x$ from f. Observe that a line of slope γ crosses the graph of f at x_0 if and only if a corresponding horizontal line crosses the graph of $f_{-\gamma}$ at x_0.

Theorem 13.116: *The set*

$$Z = \{f \in \mathcal{C}[a,b] : f \text{ crosses no lines}\}$$

is a residual subset of $\mathcal{C}[a,b]$.

Proof. We wish to express $X \setminus Z$ as a countable union of nowhere dense sets. Observe that if $f \in X \setminus Z$, there is at least one line that crosses f at some point x. In technical language, f (or $-f$) is in one of the sets A_n defined as follows.

For each $n \in \mathbb{N}$, let A_n denote those functions $f \in \mathcal{C}[a,b]$ for which there exists $\gamma \in [-n,n]$ and $x \in [a,b]$ such that

$$f_{-\gamma}(t) \leq f_{-\gamma}(x) \text{ when } t \in [a,b] \cap (x - 1/n, x)$$

and

$$f_{-\gamma}(t) \geq f_{-\gamma}(x) \text{ when } t \in [a,b] \cap (x, x + 1/n)\,.$$

In fact, if $f \in A_N$, then f is also in A_n for all $n > N$. Note that the number n plays two roles in this definition. For a function to be in A_n there must be *at least one* line whose slope is between $-n$ and $+n$ when it crosses the function. Furthermore, $1/n$ specifies the length of an interval in which that line must stay above or below the function (before crossing it again).

Let $A = \bigcup_{n=1}^{\infty} A_n$. We show that for each $n \in \mathbb{N}$, A_n is closed and that the complement of A_n is dense. It follows that each A_n is nowhere dense and hence that A is a first-category subset of $\mathcal{C}[a,b]$.

To verify that A_n is closed, let $\{f_k\}$ be a sequence of functions in A_n such that $f_k \to f$ in the space $\mathcal{C}[a,b]$. We must show $f \in A_n$. For each $k \in \mathbb{N}$, the function f_k is a member of A_n and so there exists $\gamma_k \in [-n,n]$ and $x_k \in [a,b]$ such that

$$f_{-\gamma_k}(t) \leq f_{-\gamma_k}(x_k) \text{ when } t \in [a,b] \cap (x_k - 1/n, x_k)$$

and

$$f_{-\gamma_k}(t) \geq f_{-\gamma_k}(x_k) \text{ when } t \in [a,b] \cap (x_k, x_k + 1/n)\,.$$

By the Bolzano-Weierstrass theorem (Theorem 2.40) applied to the sequence of points $\{x_k\}$ there exists a subsequence $\{x_{k_i}\}$ of $\{x_k\}$ that converges to some point x_0 in $[a,b]$. Applying the Bolzano-Weierstrass theorem again, we obtain a subsequence of that sequence, $\{x_{k_{i_j}}\}$ such that $\{\gamma_{k_{i_j}}\}$ converges to some $\gamma_0 \in [-n,n]$.

It is easy to verify (Exercise 13.14.1) that

$$f_{-\gamma_0}(t) \leq f_{-\gamma_0}(x_0) \text{ when } t \in [a,b] \cap (x_0 - 1/n, x_0)$$

and

$$f_{-\gamma_0}(t) \geq f_{-\gamma_0}(x_0) \text{ when } t \in [a,b] \cap (x_0, x_0 + 1/n)\,.$$

Thus $f \in A_n$. Since this is true for all convergent sequences chosen from A_n we conclude that A_n is closed in $\mathcal{C}[a,b]$.

To show that A_n is nowhere dense, we verify that every open ball in $\mathcal{C}[a,b]$ contains points that do not belong to A_n. Let $B(g,\varepsilon)$ be an open ball in $\mathcal{C}[a,b]$. It is easy to visualize (though tedious to verify analytically) that we can choose an appropriate sawtooth function f with many steep teeth such that $f \in B(g,\varepsilon) \setminus A_n$.

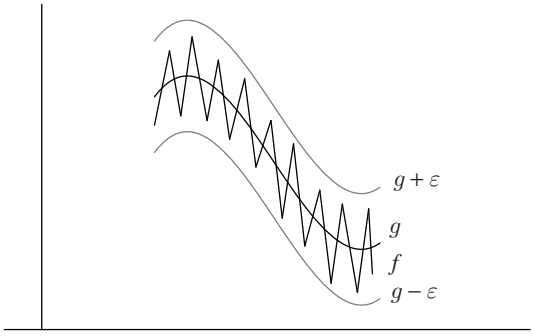


Figure 13.10. Graph of the "sawtooth" function f on an interval of length $1/n$.

Intuitively, the line segments that make up the graph of f have such steep slopes and there are so many of these segments that no line whose slope is bounded by $-n$ and n can cross the graph as required for f to be in A_n. [See Fig. 13.10. The interval depicted has length $1/n$. The slopes of the "teeth" are so large that no line with slope between $+n$ and $-n$ can cross f only once in this interval. This means that $f \in B(g,\varepsilon) \setminus A_n$.]

Thus A_n is nowhere dense, and A is of the first category. Exactly the same arguments show that the set

$$B = \{f \in C[a,b] : -f \in A\}$$

is first category. Consequently, $A \cup B$ is first category and it follows that

$$Z = C[a,b] \setminus (A \cup B)$$

is residual. It can be checked that every member of Z crosses no lines. ■

Exercises

13.14.1 Verify that

$$f_{-\gamma_0}(t) \leq f_{-\gamma_0}(x_0)$$

when $t \in [a,b] \cap (x_0 - 1/n, x_0)$ and that

$$f_{-\gamma_0}(t) \geq f_{-\gamma_0}(x_0)$$

when $t \in [a,b] \cap (x_0, x_0 + 1/n)$ in the proof of Theorem 13.116.

13.14.2 Show that every $f \in Z$ (see the proof of Theorem 13.116) has $+\infty$ and $-\infty$ as derived numbers at every point.

13.14.2 Nowhere Monotonic Functions

All of the continuous functions that we encounter in a calculus class are monotonic or else they are piecewise monotonic: They are nondecreasing on some intervals and nonincreasing on remaining intervals. Indeed it is extremely hard to imagine a continuous function that behaves differently than this. Is it possible to prove the existence of continuous *nowhere monotonic* functions in the sense of the following definition?

Definition 13.117: A function f on an interval $[a,b]$ is said to be *nowhere monotonic* if there is no subinterval $[c,d] \subset [a,b]$ on which f is monotonic.

Theorem 13.118: *The class of continuous, nowhere monotonic functions forms a residual subset of $C[a,b]$.*

Proof. When a continuous function f is monotonic on an interval $[c,d]$, it is clear that there are many lines that cross f. In fact, every horizontal line $y = k$ for k between $f(c)$ and $f(d)$ must cross f. Thus, any member of the class of functions Z discussed in the preceding section (Theorem 13.116) cannot be monotonic on any interval. Thus this theorem follows directly from Theorem 13.116 since the set of nowhere monotonic functions contains the residual set Z and hence must also be residual. ■

13.14.3 Continuous Nowhere Differentiable Functions

We have proved now, using the line crossing arguments together with Baire category arguments, the existence of continuous functions having geometric properties that are hard to visualize. Let us turn to an analytic property that is equally hard to visualize.

It is easy to construct a function that has no derivative anywhere: Simply take a function that is discontinuous everywhere. But can a continuous function exist that has no derivative anywhere? The function $f(x) = |x|$ illustrates how to arrange for at least one point of nondifferentiability. Any finite number of points can be just as easily handled. In the early nineteenth century a number of mathematicians were quite convinced that not much more could be said. Indeed some of them "proved" that a continuous function would have to have points of differentiability.

By the middle of the nineteenth century many mathematicians were aware of the existence of continuous functions that had no points of differentiability. Constructions of such functions involved summations of infinite series whose successive terms contributed increasingly to the nondifferentiability of their sum. One of the best known of such constructions was given by Weierstrass around 1875. Use of the Baire category theorem to prove the existence of such functions had to wait until 1931, at which time two Polish mathematicians Stefan Banach (1892–1945) and Stefan Mazurkiewicz (1888–1945), in separate papers in the journal *Studia Mathematica*, provided such proofs.

We call a function *nowhere differentiable* if it has a finite derivative at no point.

Theorem 13.119: *The class of continuous, nowhere differentiable functions forms a residual subset of $C[a,b]$.*

Proof. Observe that if a continuous function is differentiable at a point $x_0 \in (a,b)$, with $f'(x_0) = \gamma$, then any line whose slope is not γ and that passes through $(x_0, f(x_0))$ will cross f. This implies that each member of the class of functions Z discussed in Theorem 13.116 is differentiable at no point. Thus this theorem follows directly from Theorem 13.116 since the set of nowhere differentiable functions contains the residual set Z and hence must also be residual. ■

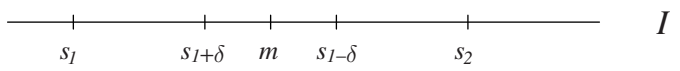


Figure 13.11. The interval I in the proof that most members of $\mathcal{K}$ are nowhere dense.

Note. In fact, a more careful analysis shows that for each $f \in Z$, $+\infty$ and $-\infty$ are derived numbers of f at every point. (See Section 7.8 for a discussion of Dini derivatives and derived numbers.) We leave this analysis as Exercise 13.14.2.

13.14.4 Cantor Sets

We turn now to an example in which the objects are closed sets. Recall that the Cantor ternary or "middle-third set" was the first example of a nonempty, nowhere dense set of real numbers without isolated points. Such sets were extremely hard to visualize by the nineteenth-century mathematicians, and many mistakes were made before Cantor's construction was known. In honor of Cantor let us call such sets by his name.

Definition 13.120: A nonempty, nowhere dense compact set of real numbers without isolated points is said to be a *Cantor set*.

Suppose that we, like the nineteenth-century mathematicians, had not heard of Cantor sets, but, unlike nineteenth-century mathematicians, did know the Baire category theorem and did know that the space $\mathcal{K}$ of closed subsets of $[0,1]$ with the Hausdorff metric d is complete (Exercises 13.8.13 and 13.3.9). We might ask, "What do most nonempty closed subsets of $[0,1]$ look like?"

> **Question 1.** Do most nonempty closed subsets of $[0,1]$ contain interior points?

If a closed set has interior points, there exists a closed interval I with rational endpoints contained in that set. Thus every closed set that does contain interior points is in the family

$$\{K \in \mathcal{K} : K \supset I\}$$

for some such interval I. Take any $F \in \mathcal{K}$ and let $\varepsilon > 0$. By Exercise 13.7.3 (and its hint) there exists a finite set S of rational numbers such that $d(S,F) < \varepsilon$. We can choose S so that $S \cap I$ has at least two members. Let $S = \{s_1, \ldots, s_n\}$ with s_1 and s_2 in I and no other points of S in between s_1 and s_2. Let δ be smaller than $|s_1 - s_2|/3$ and also sufficiently small that

$$B(S,\delta) \subset B(F,\varepsilon).$$

If $T \in B(S,\delta)$, then T cannot contain the point $m = (s_1 + s_2)/2$ (Fig. 13.11). Thus T cannot contain I. Thus the ball $B(F,\varepsilon)$ contains the ball $B(S,\delta)$, no point of which is in the set

$$\{K \in \mathcal{K} : K \supset I\}.$$

It follows that this latter set is nowhere dense in $\mathcal{K}$ for any choice of interval I.

Now let $\{I_k\}$ be an enumeration of the closed subintervals of $[0,1]$ with rational endpoints. For each $k \in \mathbb{N}$ let

$$A_k = \{K \in \mathcal{K} : K \supset I_k\}.$$

We have shown that each of the sets A_k is nowhere dense in $\mathcal{K}$; thus

$$A = \bigcup_{k=1}^{\infty} A_k$$

is a first-category set. This shows that

$$\{K \in \mathcal{K} : K \text{ has nonempty interior}\}$$

is of the first category. Thus we have answered our first question and can assert that

Most members of $\mathcal{K}$ are nowhere dense. (47)

We next ask about isolated points.

Question 2. Do most members of $\mathcal{K}$ have isolated points?

If $K \in \mathcal{K}$ has an isolated point x, then there exists $n \in \mathbb{N}$ such that

$$\operatorname{dist}(x, K \setminus \{x\}) \geq 1/n.$$

(See Exercise 13.6.11 for the definition of the function "dist.") Let

$$B_n = \{K \in \mathcal{K} : \exists\, x \in K \text{ for which } \operatorname{dist}(x, K \setminus \{x\}) \geq 1/n\}.$$

We verify easily (Exercise 13.14.3) that each of the sets B_n is nowhere dense in $\mathcal{K}$. Let $B = \bigcup_{n=1}^{\infty} B_n$. Then B is of the first category in $\mathcal{K}$. If K has any isolated points, then $K \in B$. Thus we have answered our second question and can assert that

Most members of $\mathcal{K}$ have no *isolated points.* (48)

Combining (47) and (48), we see that we have proved the following theorem.

Theorem 13.121: *Let $\mathcal{K}$ be the metric space of nonempty closed subsets of $[0,1]$ with the Hausdorff metric, and let $\mathcal{S}$ consist of the Cantor sets in $[0,1]$. Then $\mathcal{S}$ is residual in $\mathcal{K}$.*

Now for nineteenth-century mathematicians (and for students of elementary calculus), closed sets in $[0,1]$ were/are rather simple. They consist of intervals, isolated points, and limits of isolated points. With a little more effort, perhaps sets containing limits of limit points, or limits of limits of limit points could be imagined (Exercise 4.2.24). Our theorem shows that, in fact, *most* closed sets are Cantor sets, when "most" is interpreted in terms of category.

Exercises

13.14.3 Verify that the sets B_n that appear in the arguments leading to a proof of Theorem 13.121 are nowhere dense in $\mathcal{K}$.

13.14.4 Show that

$$Y = \{f \in M[a,b] : f \text{ is one-to-one}\}$$

is a residual subset of the space $M[a,b]$ of bounded functions on $[a,b]$.
SEE NOTE 376

13.14.5 Show that

$$Z = \{f \in M[a,b] : \text{ the range of } f \text{ is nowhere dense}\}$$

is a residual subset of the space $M[a,b]$ of bounded functions on $[a,b]$.

13.15 Challenging Problems for Chapter 13

13.15.1 Let X be a metric space and let $\mathcal{A}$ be a family of closed subsets of X such that if $A, B \in \mathcal{A}$, then $A \subset B$ or $B \subset A$. Let

$$E = \bigcup_{A \in \mathcal{A}} A.$$

Prove that E can be expressed as a countable union of closed subsets of X.
SEE NOTE 377

13.15.2 By a power of a map A we mean a composition with itself; thus

$$A^2(x) = A(A(x)),\ A^3(x) = A(A(A(x))),\ \ldots$$

are powers of A.

(a) Show that a map A defined on a complete metric space (X,d) for which one of the powers of A is a contraction has a unique fixed point.

(b) Show that the integral equation

$$f(x) = \lambda \int_a^x K(x,y) f(y)\, dy + \phi(x)$$

has a unique solution in $\mathcal{C}[a,b]$. (Here λ, f, K, and ϕ are as in Example 13.81.)

SEE NOTE 378

13.15.3 Prove that $[0,1]$ cannot be expressed as a countable disjoint union of closed sets (except in the obvious way as a union of one set!).
SEE NOTE 379

13.15.4 In a general metric space (X,d) take an arbitrary set $A \subset X$ and perform a sequence of complements or closures. How many distinct sets can arise in this way?
SEE NOTE 380

13.15.5 We know from elementary calculus that if a function $f : [0,1] \to \mathbb{R}$ has an nth derivative that is identically zero, then f is a polynomial on $[0,1]$. Prove that if f has derivatives of all orders on $[0,1]$ and for each $x \in [0,1]$ there exists $n(x) \in \mathbb{N}$ such that $f^{(n(x))}(x) = 0$, then f is a polynomial on $[0,1]$.
SEE NOTE 381

13.15.6 Let (X,d) be a metric space and let $f : X \to X$ be an isometry, that is, so that $d(f(x), f(y)) = d(x,y)$ for all $x, y \in X$. Show that, in general, f need not be onto but must be if X is compact.
SEE NOTE 382

13.15.7 Suppose that every real-valued continuous function defined on a metric space X attains a maximum value. Show that X must be compact.

13.15.8 Suppose that every real-valued continuous function defined on a metric space X is uniformly continuous. Show that X must be complete but need not be compact.

13.15.9 Let (X,d) be a metric space. Show that (X,d) is compact if and only if for every equivalent metric ρ on X, the space (X,ρ) is complete.
SEE NOTE 383

13.15.10 (**Liouville numbers**) This example shows that a set can be small in the sense of measure, yet large in the sense of category. A real number z is called a *Liouville number* if z is irrational and has the property that for each n there exist integers p and $q > 1$ such that

$$\left| z - \frac{p}{q} \right| < \frac{1}{q^n}.$$

Prove the following statements about the set L of Liouville numbers [named after Joseph Liouville (1809–1882)].

(a) The set of Liouville numbers can be expressed as

$$L = (\mathbb{R} \setminus \mathbb{Q}) \cap \bigcap_{n=1}^{\infty} G_n,$$

where G_n are open sets defined as

$$G_n = \bigcup_{q=2}^{\infty} \bigcup_{p=-\infty}^{\infty} \left(\frac{p}{q} - \frac{1}{q^n}, \frac{p}{q} + \frac{1}{q^n} \right).$$

(b) L is a dense set of type $\mathcal{G}_\delta$, so L is large in the sense of category (see Exercise 13.13.15).

(c) L has measure zero and so is small in the sense of measure.

13.15.11 Use Exercise 13.14.4 to show that every bounded function on an interval $[a,b]$ is the sum of two bounded one-to-one functions on $[a,b]$.
SEE NOTE 384

13.15.12 Let $X = (0,1)$. Let $d(x,y) = |x-y|$ and let

$$e(x,y) = |x-y| + \left| \frac{1}{x} - \frac{1}{y} \right|.$$

(a) Prove that e is a metric on X.

(b) Prove that a sequence $\{x_k\}$ converges in (X,e) if and only if it converges in (X,d).

(c) Prove that a set $A \subset X$ is open in (X,e) if and only if it is open in (X,d).

(d) Prove that the identity map

$$h : (X,d) \to (X,e)$$

is a homeomorphism.

(e) Prove that (X,e) is complete.

(f) Consider a Cauchy sequence in (X,d) that does not converge. It can't converge in (X,e) either. So how can (X,e) be complete?

(g) Verify that the Baire category theorem holds in (X,d) even though the space is not complete.

SEE NOTE 385

13.15.13 The space (X,d) in Exercise 13.15.12 is not complete, but that exercise shows that it can be *remetrized* as (X,e) to be complete. This means that d and e are equivalent metrics and (X,d) and (X,e) are topologically equivalent (homeomorphic) metric spaces. Such a space is said to be *topologically complete*.

(a) Verify that the Baire category theorem holds in every topologically complete metric space.

(b) Give an example of a metric space that is not topologically complete.

13.15.14 **(Urysohn's Lemma)** Prove the following theorem, which is due to Pavel Urysohn (1898–1924).

> *Let A and B be disjoint nonempty closed subsets of a metric space (X,d). Then there exists a continuous function $g : X \to \mathbb{R}$ such that $g(x) = 0$ for all $x \in A$, $g(x) = 1$ for all $x \in B$ and $0 < g(x) < 1$ for all $x \in X \setminus (A \cup B)$.*

SEE NOTE 386

13.15.15 This problem requires a bit of knowledge about complex numbers and transcendental numbers. We exhibit two sets R and T in $\mathbb{R}^2$ such that R and T are congruent and each is congruent to their union $S = R \cup T$.

For each complex number z, let $t(z) = z+1$, $r(z) = e^i z$. Thus t is just a right translation by 1 unit and r is a rotation by 1 radian. Let S consist of those points that can be obtained by a finite number of applications of t and r starting from the origin. Each member of S can be represented as a polynomial in e^i with positive integer coefficients. (For example, if we translate five times, then rotate twice, then translate once more, the resulting point can be represented as $5e^{2i} + 1$.) Since e^i is transcendental, the representation is unique.

Let R consist of those points that have no constant term in their representation, and let $T = S \setminus R$. Prove that $t(S) = T$ and $r(S) = R$ so R, T and $S = R \cup T$ are pairwise isometric. Note that the isometries involved are isometries of $\mathbb{R}^2$ onto $\mathbb{R}^2$, not just isometries among the sets R, S, and T.

13.15.16 Let (X,d) be a metric space and let $M(X)$ denote the set of all bounded real-valued functions on X furnished with the sup metric. Choose a fixed $x_0 \in X$ and define a mapping $h : X \to M(X)$ by writing $h(x)$ for the function

$$(h(x))(y) = d(x,y) - d(y,x_0) \quad (y \in X).$$

(a) Show that each $h(x)$ is a bounded function on X for each $x \in X$.

(b) Show that h is an isometry of X to a subspace of $M(X)$.

(c) Show that *every* metric space is isometric to a subspace of some complete function space (i.e., a complete metric space of real-valued functions defined on a set and furnished with the sup metric).

Notes

[288]Exercise 13.1.2. The vector space axioms are taught in any elementary linear algebra course or text. We have used them in Theorem 11.1 to describe the vector space structure of $\mathbb{R}^n$. Note that, in this example, (d) is a special case of (b), but, in general, it is not true that scalar multiplication is a special case of multiplication of elements of the space. (Here, scalars can be viewed as constant functions.)

[289]Exercise 13.1.4. For example, define $f \preceq g$ if $g - f$ is both nonnegative and differentiable on $[0,1]$.

[290]Exercise 13.2.1. The first two are not metrics. The metrics in (d) and (e) are interesting since they are bounded, that is, $d(x,y) \leq 1$ for all x, y, but nonetheless these metrics are closely related to the usual metric: Open sets, closed set, convergent sequences etc. are the same under these metrics. To check the triangle inequality in (e) check first that the function $f(t) = t/(1+t)$ is increasing on $[0,\infty)$. Then, use the fact that

$$f(|x-y|) \leq f(|x-y|+|y-z|).$$

Finally, note that (f) is the discrete metric.

[291]Exercise 13.2.7. This idea was already used for the metric $d(x,y) = |x-y|$ in $X = \mathbb{R}$. See Exercise 13.2.1.

[292]Exercise 13.3.5. Check that all the properties of a metric hold except for condition (2).

[293]Exercise 13.3.6. While there is a subset relation here, there is no subspace relation because the metrics in the three spaces do not agree.

[294]Exercise 13.3.7. For this exercise we have merely to determine which of the sets given is a *subset* of $M(\mathbb{R})$. (Which of these classes might contain an unbounded function?)

[295]Exercise 13.4.1. Note that $d(x_n,x) < 1$ if and only if $x_n = x$.

[296]Exercise 13.4.2. Use the same ideas and methods as we used in Example 13.16.

[297]Exercise 13.4.3. The notation

$$x^{(n)} = (x_1^{(n)}, x_2^{(n)}, x_3^{(n)}, \dots)$$

might be a bit confusing at first. We have here a sequence of points $x^{(1)}$, $x^{(2)}$, $x^{(3)}$, $x^{(4)}$, ... in the space ℓ_2, each of which is, in turn, a sequence of real numbers. Only one direction in the exercise is true. Show that the condition is necessary but not sufficient. Find a counterexample with

$$x^{(n)} = (x_1^{(n)}, x_2^{(n)}, x_3^{(n)}, \dots)$$

converging coordinate-wise to zero but not convergent in the metric.

[298]Exercise 13.4.5. This is false. Construct a sequence of continuous functions with $d(f_n,0) \to 0$ but not converging pointwise to the zero function.

[299]Exercise 13.4.6. The proofs of (a) and (b) can be obtained by copying fairly closely the corresponding proofs for real sequences. There is also an advantage here for us that we can use the theory of real sequences. Thus to prove (a) you can derive it directly from the inequality

$$d(x,y) \leq d(x,x_n) + d(y,x_n).$$

For further generalizations be careful: A general metric space has no addition, multiplication, or order. Even if a metric space *does* have some other structure, such as addition, it would still be necessary to assume some special properties of the metric on the space in order to obtain properties such as that

$$\lim_{n\to\infty} x_n + y_n = \lim_{n\to\infty} x_n + \lim_{n\to\infty} y_n.$$

[300]Exercise 13.4.9. Show that $t/(1+t) \leq t$ for all $t \geq 0$. From this deduce that convergence in d implies convergence in e_2. Show that

$$t/4 \leq t/(1+t)$$

for

$$0 \leq t \leq 1.$$

From this deduce that convergence in e_2 implies convergence in d.

[301]Exercise 13.4.10. Use the metrics from Exercises 13.2.7 and 13.2.8.

[302]Exercise 13.4.12. If (a) holds, then $x_n \to x$ in (X,d_2) implies that $x_n \to x$ in (X,d_1) but not necessarily conversely. A similar statement is true for (b). If (c) holds, then $x_n \to x$ in (X,d_1) if and only if $x_n \to x$ in (X,d_2).

[303]Exercise 13.4.14. For (a) part your answer will depend on $[a,b]$. For example, if $[a,b] = [0,1/2]$, then the answer is that it does converge, while if $[a,b] = [0,1]$, then it does not.

[304]Exercise 13.4.17. It is true that a sequence $\{p_n\}$ of polynomials converges in $\mathcal{P}[a,b]$ if and only if it converges uniformly *to a polynomial*.

[305]Exercise 13.4.18. For the converse show that $x_n = 1/n$ is a Cauchy sequence in the metric space $X = (0,1)$ considered as a subspace of $\mathbb{R}$ with the usual metric.

[306]Exercise 13.5.4. For the counterexample, take the space as $\mathbb{R}^2$ and E and F are the graphs of $f(x) = e^x$ and $g(x) = -e^x$, respectively.

[307]Exercise 13.5.7. Consider the discrete space. (For unusual examples this is always worth a try first.)

[308]Exercise 13.5.12. This will require the application of some of the theorems of Chapter 9 since convergence in this space is exactly uniform convergence.

[309]Exercise 13.5.15. While it might seem plausible that G should be open, it is not. Show that G contains no ball $B(x_0,r)$ for

$$x_0 = (0,0,0,\dots)$$

and any $r > 0$.

[310]Exercise 13.5.19. An account can be found in Theorem 11.19.

[311]Exercise 13.5.20. A proof of the Heine-Borel theorem in $\mathbb{R}^2$ can be based on the Bolzano-Weierstrass theorem (Exercise 13.5.19). First show that from an open cover of any closed and bounded subset $K \subset \mathbb{R}^2$ you can extract a sequence G_1, G_2, G_3, ... of open sets covering K. Then show that for some N the sets G_1, G_2, G_3, ..., G_N must cover K otherwise there is a sequence of points in K that has no convergent subsequence.

[312]Exercise 13.6.1. Show for the first question that all functions are continuous and for the other that only constant functions are continuous.

[313]Exercise 13.6.7. Calculate $f(x,x^2)$ and use this to find a sequence of points $(u_n,v_n) \to 0$ so that $f(u_n,v_n)$ does not converge to 0.

[314]Exercise 13.6.8. Write out the ε-δ statement that expresses the statement that "for each $x_2 \in \mathbb{R}$ the function $x \to f(x,x_2)$ is continuous." What would be the uniform version of that? For further discussion of these ideas, see Exercise 11.7.13.

[315]Exercise 13.6.11. For (g) use the function

$$\operatorname{dist}(x,E)[\operatorname{dist}(x,E) + \operatorname{dist}(x,F)]^{-1}.$$

Deduce (h) from (g). [For (h) do not be tempted to think that E and F must be a positive distance apart.]

[316]Exercise 13.6.12. Continuous curves don't always look like ones we've seen. In 1890 the Italian mathematician Giuseppe Peano (1858–1932) gave an example of a continuous curve that fills the unit square $[0,1]\times[0,1]$. Hausdorff, in 1914, stated that Peano's result was "one of the most remarkable results of set theory." Note that a continuous curve has to be continuous as a function from $[0,1]$ to $\mathbb{R}^2$ but does not have to be one-to-one (i.e., it can cross itself). It can be shown that a one-to-one continuous curve could not fill the unit square.

[317]Exercise 13.6.15. You can assume and use Bessel's inequality (given in the example). F. W. Bessel (1784–1846) is credited with this elementary and easy to prove inequality. His name is more famously attached to a special class of functions that have become an indispensable tool in applied mathematics, physics, and engineering. The interest in these functions arose in the treatment of the problem of the perturbation in the planetary system.

[318]Exercise 13.6.18. A closed and bounded subset $K\subset\mathbb{R}^2$ has the same properties that a closed and bounded set of real numbers has, namely the compactness properties of Section 4.5. (See Exercise 13.5.19 or Exercise 13.5.20.) Thus you can use similar proofs to those of Exercise 5.6.12 (Bolzano-Weierstrass property) or Exercise 5.6.13 (Heine-Borel property) to prove uniform continuity of continuous functions on closed and bounded sets. For (c) and (d) also imitate the proof for functions of one variable. (See also the discussion in Sections 11.8 and 11.9.)

[319]Exercise 13.6.26. See Example 13.36 for a homeomorphism of $\mathbb{R}$ and $(-1,1)$. A similar idea will show that $\mathbb{R}$ and $(0,1)$ are also homeomorphic. The Bolzano-Weierstrass theorem will help show that $\mathbb{R}$ and $[0,1]$ are not homeomorphic: Any sequence in $[0,1]$ would have to have a convergent subsequence.

[320]Exercise 13.6.27. If $h:[0,\infty)\to\mathbb{R}$ is a homeomorphism, show that h is either increasing or decreasing.

[321]Exercise 13.6.28. Compare with Exercise 13.2.7.

[322]Exercise 13.6.29. The answer is no for the converse. Show that the subspace

$$X=\{1,1/2,1/3,1/4,\dots\}$$

of $\mathbb{R}$ with the usual metric is topologically equivalent to a discrete space but that

$$\inf\{|x-y|:x,y\in X,\ x\neq y\}=0.$$

[323]Exercise 13.6.37. See also the discussion of connected sets in $\mathbb{R}^n$ given in Section 11.10.

[324]Exercise 13.6.38. Use Exercise 13.6.11(d) to obtain, if X contains at least two points, a continuous real-valued function on X that is not constant. Explain why the range of this function cannot be finite or countable.

[325]Exercise 13.6.40. Even for subspaces of $\mathbb{R}^2$ the converse fails. Use the set

$$\{(x,\sin(1/x):0<x<1\}$$

$$\cup\{(0,x):-1\le x\le 1\}.$$

[326]Exercise 13.6.41. What set in $\mathbb{R}$ would map onto the x-axis in $\mathbb{R}^2$?

[327]Exercise 13.6.42. Consider the function

$$h:K\times K\to K$$

defined by

$$h(a,b) = .a_1b_1a_2b_2\ldots,$$

where $a = .a_1a_2\ldots$, $b = .b_1b_2\ldots$ are appropriate base 3 representations of $a,b \in K$.

[328]Exercise 13.6.43. For (c) let

$$h(\{a_1,a_2,a_3,\ldots\}) = .b_1b_2b_3\ldots,$$

where $b_k = 2a_k$.

[329]Exercise 13.6.44. For (b) contrast the case that the limit point is unilateral (or bilateral) in both X and Y with the case that one is a unilateral limit point and the other is a bilateral point.

[330]Exercise 13.6.47. Give an example of two subsets A and B of $\mathbb{R}$, each of which is isometric to a subset of the other but that are not themselves isometric. Use

$$A = \{2,3,4,\ldots\}$$

and

$$B = \{0\} \cup \{2,3,4,\ldots\}.$$

[331]Exercise 13.6.52. Note that $(0,1) \subset X$ would map onto a set of diameter 1.

[332]Exercise 13.7.3. The family of finite sets of rational numbers in $[0,1]$ forms a countable dense subset of $\mathcal{K}$.

[333]Exercise 13.7.4. By a polygonal function on $[a,b]$ we mean a continuous, piecewise linear function, the corners (the points at which the right- and left-hand derivatives are different) are called the vertices. Here both coordinates of any vertex are assumed to be rational numbers. Use uniform continuity to show that each function in $\mathcal{C}[a,b]$ can be approximated by such a function arbitrarily closely. Now show that the set of such functions is countable.

[334]Exercise 13.7.5. Here is a warning: The countable dense subset known to exist for the space might not be a subset of the subspace.

[335]Exercise 13.7.8. Several of the spaces we have considered are not separable and can be used for (b).

[336]Exercise 13.7.11. We are trying to find, if possible, an embedding of (X,d) into these spaces. You should be able to use the separability of $\mathcal{C}[0,1]$ to determine whether this is possible. For the other two Examples 13.54 and 13.55 should assist in the construction of the embedding.

[337]Exercise 13.8.1. This follows easily from the triangle inequality.

[338]Exercise 13.8.2. The methods used to prove that convergent sequences of real numbers are bounded in Theorem 2.11 can be imitated here.

[339]Exercise 13.8.3. If $\{x_n\}$ is Cauchy, then show it is true that

$$\lim_{n\to\infty} d(x_n,x_{n+1}) = 0.$$

The converse is false. For example, the sequence $x_n = \sqrt{n}$ in $\mathbb{R}$ has the property that

$$\lim_{n\to\infty} d(x_n,x_{n+1}) = 0$$

but it is not Cauchy (nor even bounded).

[340]Exercise 13.8.4. Using the triangle inequality, show that the sequence $\{d(x_n,y_n)\}$ is a Cauchy sequence of real numbers.

[341]Exercise 13.8.5. For each positive $\varepsilon_k = 2^{-k}$, show that there is an integer n_k with

$$d(x_m, x_n) < \varepsilon_k$$

for $m, n \geq n_k$.

[342]Exercise 13.8.6. The methods used to prove this assertion for real numbers in Theorem 2.41 can be imitated here.

[343]Exercise 13.8.7. In one direction check that a sequence with the property that

$$\sum_{k=1}^{\infty} d(x_k, x_{k+1})$$

converges is Cauchy. In the other direction note that a Cauchy sequence need not have this property, but it must have a subsequence with that property. (To construct the subsequence Exercise 13.8.5 should help; then use Exercise 13.8.6.)

[344]Exercise 13.8.8. In one direction use the sequence of closed balls to select a Cauchy sequence. In the other direction, if $\{x_n\}$ is Cauchy you should be able to construct an appropriate sequence of closed balls with the help of Exercise 13.8.5.

[345]Exercise 13.8.13. (a) Show first that if $\{H_n\}$ is a decreasing sequence of nonempty closed sets in $[0,1]$ and $H = \bigcap_1^\infty H_n$, then $H_n \to H$ in $\mathcal{K}$. (b) Show that if $\{A_n\}$ is a Cauchy sequence in $\mathcal{K}$ and H_n denotes the closure of the set

$$\bigcup_{k=n}^{\infty} A_k,$$

then $\{H_n\}$ is a decreasing sequence of closed sets,

$$H = \bigcap_{1}^{\infty} H_n$$

is a nonempty closed set, and $H_n \to H$. (c) Finally, show $A_n \to H$.

[346]Exercise 13.8.16. It is important to realize that completeness is not a topological property. Consider $X = \mathbb{R}$ and $Y = (-1,1)$ and review Example 13.36.

[347]Exercise 13.8.18. Use Theorem 13.66.

[348]Exercise 13.9.8. A is not a contraction (in contrast to Example 13.77), but A^2 is. The unique fixed point is easy to find.

[349]Exercise 13.9.9. Show that

$$d(a,b) < \frac{\varepsilon}{1-\alpha},$$

where α has the property for all $x, y \in X$ either

$$d(A(x), A(y)) \leq \alpha d(x,y)$$

or

$$d(B(x), B(y)) \leq \alpha d(x,y).$$

Use the technique in the proof of Theorem 13.75 and observe that

$$d(a, B(a)) = d(A(a), B(a)) < \varepsilon.$$

[350]Exercise 13.9.13. Compare this with Exercises 13.9.11 and 13.9.12, where, contrary to this exercise, the mappings A_n are assumed to be contractions. Check the fact that

$$d(a_n, A(a_n) = d(A_n(a_n), A(a_n)) \to 0.$$